

# TOPICS IN MATHEMATICS SCIENCE VII

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## INTRODUCTION TO KOSZUL DUALITY

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### Convention

Throughout the course,  $\mathbb{k}$  is a field. All rings are unital and associative. All modules are right modules unless otherwise specified. When we discuss dimensions, we mean dimensions as  $\mathbb{k}$ -vector spaces.

Indented passages with a pale beige background and a dark green left rule contain advanced material for readers seeking greater depth; beginners may omit them.

## 1 Graded algebras, modules, and duals

### 1.1 Algebras and modules

**Definition 1.1.** A  $\mathbb{k}$ -algebra is a  $\mathbb{k}$ -vector space with an associative unital multiplication that is *bilinear*, meaning linear in each variable.

A right  $A$ -module is understood to be a  $\mathbb{k}$ -vector space  $M$  with a bilinear action satisfying  $(ma)b = m(ab)$  and  $m1_A = m$  for  $m \in M$  and  $a, b \in A$ . A module homomorphism  $f : M \rightarrow N$  is linear and satisfies  $f(ma) = f(m)a$ . A homomorphism of  $\mathbb{k}$ -algebras is a linear map preserving multiplication and the identity.

**Example 1.2.** (1) The matrix ring  $\text{Mat}_n(\mathbb{k})$  is a  $\mathbb{k}$ -algebra. The space of  $1 \times n$  row vectors is a right  $\text{Mat}_n(\mathbb{k})$ -module by matrix multiplication.

(2) Let  $\mathbb{k}A_n$  be the *upper triangular matrix ring*, which is also a  $\mathbb{k}$ -algebra.

$$\mathbb{k}A_n = \begin{pmatrix} \mathbb{k} & \mathbb{k} & \cdots & \mathbb{k} \\ 0 & \mathbb{k} & \cdots & \mathbb{k} \\ 0 & 0 & \ddots & \vdots \\ 0 & \cdots & 0 & \mathbb{k} \end{pmatrix}$$

For those familiar with path algebras,  $\mathbb{k}A_n$  is the path algebra of a *linearly oriented  $A_n$  quiver*, which explains the notation. We use matrix calculations, so no knowledge of quivers is required. Write  $e_{ij}$  for the matrix with 1 in entry  $(i, j)$  and zero elsewhere. Our convention composes paths from left to right:  $e_{ij}$  represents  $i \rightarrow j$ , and  $e_{ij}e_{j\ell} = e_{i\ell}$ .

## 1.2 Categories and functors

We organise modules together with their homomorphisms. A category records both: specifying the objects alone does not specify a category.

**Definition 1.3.** A *category*  $\mathcal{C}$  consists of objects, a set  $\text{Hom}_{\mathcal{C}}(X, Y)$  of *morphisms*  $f : X \rightarrow Y$  for each pair of objects, and composition sending  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$  to  $g \circ f : X \rightarrow Z$ . Each object  $X$  has an identity  $\text{id}_X : X \rightarrow X$ , and

$$h \circ (g \circ f) = (h \circ g) \circ f, \quad \text{id}_Y \circ f = f = f \circ \text{id}_X \quad \text{for all composable } f, g, h.$$

A morphism  $f : X \rightarrow Y$  is an *isomorphism* if some  $g : Y \rightarrow X$  satisfies  $g \circ f = \text{id}_X$  and  $f \circ g = \text{id}_Y$ ; we then write  $X \cong Y$ . A *full subcategory* of  $\mathcal{C}$  consists of chosen objects of  $\mathcal{C}$ , with all morphisms between them and the same composition.

**Example 1.4.** For a  $\mathbb{k}$ -algebra  $A$ , we use the notation

$$\begin{aligned} \text{Mod } A &:= \text{category of right } A\text{-modules and } A\text{-linear maps,} \\ \text{mod } A &:= \text{full subcategory of finitely generated right } A\text{-modules.} \end{aligned}$$

In either category,  $\text{Hom}_A(M, N)$  is the space of  $A$ -linear maps  $M \rightarrow N$ ; composition is the usual composition of maps. For  $A = \mathbb{k}$ , these are the categories of all and of finite-dimensional vector spaces, respectively. More generally, if  $\dim_{\mathbb{k}} A < \infty$ , an  $A$ -module is finitely generated if and only if it is finite-dimensional: a finite generating set gives a quotient of  $A^{\oplus n}$ , and a finite  $\mathbb{k}$ -basis is also an  $A$ -generating set.

**Definition 1.5.** A *functor*  $F : \mathcal{C} \rightarrow \mathcal{D}$  assigns an object  $F(X)$  to each object  $X$  and a morphism  $F(f) : F(X) \rightarrow F(Y)$  to each  $f : X \rightarrow Y$ , such that

$$F(\text{id}_X) = \text{id}_{F(X)}, \quad F(g \circ f) = F(g) \circ F(f) \quad \text{for all composable } f, g.$$

For example, forgetting the  $A$ -action gives a functor  $\text{Mod } A \rightarrow \text{Mod } \mathbb{k}$ : it keeps the underlying vector spaces and linear maps. The *identity functor*  $\text{id}_{\mathcal{C}}$  leaves every object and morphism unchanged; functors compose by applying one after the other. Every functor preserves isomorphisms, since  $F(f^{-1})$  is inverse to  $F(f)$ .

The *opposite category*  $\mathcal{C}^{\text{op}}$  has the same objects as  $\mathcal{C}$ , with  $\text{Hom}_{\mathcal{C}^{\text{op}}}(X, Y) = \text{Hom}_{\mathcal{C}}(Y, X)$  and composition reversed. A *contravariant functor* from  $\mathcal{C}$  to  $\mathcal{D}$  means a functor  $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ . Thus it sends  $f : X \rightarrow Y$  to  $F(f) : F(Y) \rightarrow F(X)$  and satisfies  $F(g \circ f) = F(f) \circ F(g)$ . Duals will provide our main examples.

**Definition 1.6.** For functors  $F, G : \mathcal{C} \rightarrow \mathcal{D}$ , a *natural transformation*  $\eta : F \rightarrow G$  is a family of morphisms  $\eta_X : F(X) \rightarrow G(X)$  such that

$$\begin{array}{ccc} F(X) & \xrightarrow{\eta_X} & G(X) \\ F(f) \downarrow & & \downarrow G(f) \\ F(Y) & \xrightarrow{\eta_Y} & G(Y) \end{array} \quad G(f) \circ \eta_X = \eta_Y \circ F(f) \quad \text{for every } f : X \rightarrow Y. \quad (1.1)$$

If every  $\eta_X$  is an isomorphism,  $\eta$  is a *natural isomorphism*, written  $F \cong G$ .

The square in (1.1) is said to *commute*: its two routes from  $F(X)$  to  $G(Y)$  give the same morphism. For example, for fixed  $c \in \mathbb{k}$ , the maps  $c \text{id}_M : M \rightarrow M$  define a natural transformation  $\text{id}_{\text{Mod } A} \rightarrow \text{id}_{\text{Mod } A}$  because  $f(cm) = cf(m)$  for every module homomorphism  $f : M \rightarrow N$ . It is a natural isomorphism when  $c \neq 0$ .

A functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  is an **equivalence** if there is a functor  $G : \mathcal{D} \rightarrow \mathcal{C}$  with natural isomorphisms  $GF \cong \text{id}_{\mathcal{C}}$  and  $FG \cong \text{id}_{\mathcal{D}}$ . A **duality** between  $\mathcal{C}$  and  $\mathcal{D}$  is an equivalence  $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ . Thus an equivalence recovers objects and maps up to natural isomorphism; a duality also reverses the direction of maps.

**Example 1.7 ( $\mathbb{k}$ -linear duality).** For a vector space  $V$ , define its  **$\mathbb{k}$ -linear dual** by  $D(V) := \text{Hom}_{\mathbb{k}}(V, \mathbb{k})$ . For a linear map  $u : V \rightarrow W$ , set

$$D(u) : D(W) \longrightarrow D(V), \quad \lambda \longmapsto \lambda \circ u.$$

The identities  $D(\text{id}_V) = \text{id}_{D(V)}$  and  $D(v \circ u) = D(u) \circ D(v)$  make  $D : (\text{Mod } \mathbb{k})^{\text{op}} \rightarrow \text{Mod } \mathbb{k}$  a functor. On finite-dimensional spaces, the canonical map to the bidual gives a natural isomorphism, which we denote by  $\text{coev}$ :

$$\text{coev} : \text{id}_{\text{Mod } \mathbb{k}} \xrightarrow{\sim} D \circ D, \quad \text{coev}_V : V \longrightarrow D(D(V)), \quad x \longmapsto (\lambda \mapsto \lambda(x)).$$

Indeed, if  $x \neq 0$ , some  $\lambda \in D(V)$  satisfies  $\lambda(x) \neq 0$ , so  $\text{coev}_V$  is injective; equal finite dimensions give surjectivity. Naturality follows from

$$D(D(u))(\text{coev}_V(x))(\lambda) = \lambda(u(x)) = \text{coev}_W(u(x))(\lambda) \quad (x \in V, \lambda \in D(W)).$$

We write  $\text{ev}_V := \text{coev}_V^{-1} : D(D(V)) \rightarrow V$  for the inverse:  $\text{ev}_V(\Phi)$  is the unique  $x \in V$  such that  $\Phi(\lambda) = \lambda(x)$  for every  $\lambda \in D(V)$ . Consequently,  $D : (\text{mod } \mathbb{k})^{\text{op}} \rightarrow \text{mod } \mathbb{k}$  is a duality.

### 1.3 Semisimple algebras

**Definition 1.8.** A module  $L$  is **simple** if  $L \neq 0$  and its only submodules are  $0$  and  $L$ . A module is **semisimple** if it is a direct sum of simple modules. An algebra  $B$  is **semisimple** if it is semisimple as a right module  $B_B$  over itself. The **Jacobson radical**  $\text{rad } B$  is the intersection of all maximal right ideals of  $B$ .

We use the following standard ring-theoretic facts without proof. Recall that a **division  $\mathbb{k}$ -algebra** is a non-zero  $\mathbb{k}$ -algebra whose non-zero elements are invertible.

**Theorem 1.9 (Artin–Wedderburn).** For a non-zero finite-dimensional  $\mathbb{k}$ -algebra  $B$ ,

$$B \text{ is semisimple} \iff \text{rad } B = 0 \iff B \cong \prod_{i=1}^r \text{Mat}_{n_i}(D_i),$$

where  $r, n_i \geq 1$  and each  $D_i$  is a finite-dimensional division  $\mathbb{k}$ -algebra. If  $\mathbb{k}$  is algebraically closed, then  $D_i = \mathbb{k}$  for every  $i$ . For semisimple  $B$ , every left or right  $B$ -module is semisimple, and every submodule has a complement. In particular, the definition is independent of the chosen side.

**Corollary 1.10.** Every finite-dimensional left or right module over a semisimple algebra  $S$  is a direct summand of  $S^{\oplus n}$  for some  $n$ .

**Proof** A finite generating set gives a surjection  $S^{\oplus n} \twoheadrightarrow M$ . By Theorem 1.9, its kernel has a complement, which maps isomorphically onto  $M$ .  $\square$

**Example 1.11.** The algebras  $\mathbb{k}^{\oplus r}$  and  $\text{Mat}_n(\mathbb{k})$  are semisimple by Theorem 1.9. For  $A = \mathbb{k}A_2$ , the ideal  $J = \mathbb{k}e_{12}$  satisfies  $J^2 = 0$ . If  $L$  is simple, then  $LJ$  is either  $0$  or  $L$ ; the latter would give  $L = LJ^2 = 0$ . Hence every maximal right ideal contains  $J$ . Since  $A/J \cong \mathbb{k} \oplus \mathbb{k}$ , those ideals are  $\mathbb{k}e_{11} \oplus \mathbb{k}e_{12}$  and  $\mathbb{k}e_{12} \oplus \mathbb{k}e_{22}$ . Their intersection is  $J$ , so  $\text{rad } A = \mathbb{k}e_{12} \neq 0$ .

## 1.4 Graded algebras

**Definition 1.12.** Let  $G$  be a group with identity  $e$ . A  $G$ -graded  $\mathbb{k}$ -algebra is a  $\mathbb{k}$ -algebra  $A$  with  $1_A \in A_e$  and a decomposition

$$A = \bigoplus_{g \in G} A_g \quad \text{such that} \quad A_g A_h \subset A_{gh} \quad \text{for all } g, h \in G.$$

An element of  $A_g$  is *homogeneous of degree  $g$* .

**Convention.** Unless stated otherwise, a graded algebra is  $\mathbb{Z}$ -graded and concentrated in non-negative degrees.

Note that the decomposition is into a direct sum, so every element of a graded algebra  $A$  can always be written as a *finite* sum of homogeneous elements.

**Example 1.13.** For the upper triangular matrix ring  $\mathbb{k}A_n$  of Example 1.2, the decomposition

$$(\mathbb{k}A_n)_d = \text{span}_{\mathbb{k}}\{e_{ij} \mid j - i = d\}$$

makes  $\mathbb{k}A_n$  a graded algebra.

**Example 1.14.** The algebra  $\mathbb{k}A_3$  also has a  $\mathbb{Z}^{\oplus 2}$ -grading with

$$\deg e_{ii} = (0, 0), \quad \deg e_{12} = (1, 0), \quad \deg e_{23} = (0, 1), \quad \deg e_{13} = (1, 1).$$

The homomorphism  $\mathbb{Z}^{\oplus 2} \rightarrow \mathbb{Z}$ ,  $(p, q) \mapsto p + q$ , gives the grading of Example 1.13. Alternatively, setting  $\deg e_{ij} = 2(j - i)$  gives a  $\mathbb{Z}$ -grading with  $A_1 = 0$  but  $A_2 \neq 0$ .

**Definition 1.15.** A *homomorphism of graded algebras*  $f : A \rightarrow B$  is a  $\mathbb{k}$ -algebra homomorphism with  $f(A_i) \subset B_i$  for all  $i \in \mathbb{Z}$ .

**Definition 1.16.** A graded algebra  $A$  is *locally finite* if  $\dim_{\mathbb{k}} A_i < \infty$  for every  $i \in \mathbb{Z}$ .

## 1.5 Graded modules

**Definition 1.17.** Let  $A = \bigoplus_{g \in G} A_g$  be a  $G$ -graded algebra. A  $G$ -graded right  $A$ -module is a right  $A$ -module  $M$  with a direct sum decomposition

$$M = \bigoplus_{g \in G} M_g \quad \text{such that} \quad M_g A_h \subset M_{gh} \quad \text{for all } g, h \in G.$$

An element of  $M_g$  is called *homogeneous of degree  $g$* .

A  $G$ -graded  $A$ -module  $M = \bigoplus_{g \in G} M_g$  is *locally finite* if  $\dim_{\mathbb{k}} M_g < \infty$  for all  $g \in G$ .

Unless specified otherwise, graded modules are  $\mathbb{Z}$ -graded; their degrees need not be non-negative.

The *opposite algebra*  $A^{\text{op}}$  has the same vector space as  $A$ , with multiplication  $a^{\text{op}} b^{\text{op}} = (ba)^{\text{op}}$ . A left  $A$ -module is a right  $A^{\text{op}}$ -module by  $m \cdot a^{\text{op}} = am$ . For a  $\mathbb{Z}$ -graded algebra,  $(A^{\text{op}})_i = A_i$ . Thus  $\text{Mod}(A^{\text{op}})$  is the category of left  $A$ -modules and left  $A$ -linear maps. It is not the opposite category  $(\text{Mod } A)^{\text{op}}$ : the latter keeps right modules as objects and reverses the morphisms.

**Definition 1.18.** An  *$A$ -bimodule* is a vector space  $M$  with  $\mathbb{k}$ -bilinear left and right  $A$ -module actions satisfying  $(am)b = a(mb)$  for all  $a, b \in A$  and  $m \in M$ . It is graded if  $M = \bigoplus_{j \in \mathbb{Z}} M_j$  and  $A_i M_j A_k \subset M_{i+j+k}$  for all  $i, j, k \in \mathbb{Z}$ .

The algebra  $A$  is a graded  $A$ -bimodule by multiplication, and each  $A_i$  is an  $A_0$ -bimodule. Bilinearity forces the left and right scalar actions to agree:  $(c1_A)m = cm = m(c1_A)$  for  $c \in \mathbb{k}$ . More precisely, we should have speak of ‘ $\mathbb{k}$ -central  $A$ -bimodule’ instead; but we will always work over a field  $\mathbb{k}$  and so will not make such distinction.

**Example 1.19.** The decomposition  $A = \bigoplus_{i \in \mathbb{Z}} A_i$  makes the right regular module  $A_A$  a graded module, and  ${}_A A_A$  a graded bimodule.

**Definition 1.20.** Let  $M$  be a graded  $A$ -module and let  $d \in \mathbb{Z}$ . The (degree/grading) *shift*  $M(d)$  is the graded  $A$ -module whose underlying  $A$ -module is  $M$ , but whose grading is

$$M(d)_j = M_{j+d}.$$

By Definition 1.20,  $1 \in A_0 = A(-d)_d$ , so  $A(-d)$  has a generator in degree  $d$ . If  $M$  is concentrated in degree 0, then  $M(d)$  is concentrated in degree  $-d$ .

**Example 1.21.** For  $A = \mathbb{k}A_3$  with the grading of Example 1.13, the module  $P_1 := e_{11}A$  consists of matrices supported on the first row. Its basis  $e_{11}, e_{12}, e_{13}$  has degrees 0, 1, 2, respectively; in  $P_1(-2)$ , the same basis has degrees 2, 3, 4.

**Definition 1.22.** Let  $M, N$  be graded  $A$ -modules. A homomorphism  $f : M \rightarrow N$  is *homogeneous of degree  $d$*  if  $f(M_j) \subset N_{j+d}$  for all  $j \in \mathbb{Z}$ . A *homomorphism of graded modules* means a homogeneous  $A$ -module homomorphism of degree 0, also said to *preserve degrees*.

**Example 1.23.** Let  $S = \mathbb{k}e_1 \oplus \mathbb{k}e_2$  with componentwise multiplication, concentrated in degree 0. Write

$$M = \bigoplus_{i \geq 0} (e_1 S)(-i) = \bigoplus_{i \geq 0} \mathbb{k}m_i, \quad \deg m_i = i, \quad m_i e_1 = m_i, \quad m_i e_2 = 0,$$

and let  $N = e_1 S$  be concentrated in degree 0. Define the  $S$ -linear map  $f : M \rightarrow N$  by

$$f\left(\sum_{i \geq 0} c_i m_i\right) = \sum_{c_i \neq 0} c_i e_1 \quad \text{for finitely supported } (c_i)_{i \geq 0}$$

so that  $f(m_i) = e_1 \neq 0$  for every  $i \geq 0$ . On the summand  $(e_1 S)(-i)$ , the restriction  $f|_{(e_1 S)(-i)} : (e_1 S)(-i) \rightarrow e_1 S$  is the underlying identity map, of degree  $-i$ . A homogeneous map  $M \rightarrow N$  of degree  $d$  vanishes on  $m_i$  unless  $d = -i$ . Hence  $f$  is not a finite sum of homogeneous maps.

**Example 1.24.** For  $A = \mathbb{k}A_3$ , left multiplication by  $e_{12}$  yields a map

$$\begin{aligned} e_{22}A &\longrightarrow e_{11}A, \\ \begin{pmatrix} 0 & 0 & 0 \\ 0 & u & v \\ 0 & 0 & 0 \end{pmatrix} &\longmapsto \begin{pmatrix} 0 & u & v \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (u, v \in \mathbb{k}). \end{aligned}$$

The  $u$  and  $v$  entries have degrees 0, 1 in the source and 1, 2 in the target, and so the map has degree 1.

**Notation.** For graded right  $A$ -modules  $M, N$ , the space of homomorphisms preserving degrees is

$$\mathrm{Hom}_A^{\mathbb{Z}}(M, N) = \{f \in \mathrm{Hom}_A(M, N) \mid f(M_j) \subset N_j \text{ for all } j \in \mathbb{Z}\}. \quad (1.2)$$

Thus  $\mathrm{Hom}_A^{\mathbb{Z}}(M, N(d))$  consists of homomorphisms  $M \rightarrow N$  of degree  $d$ , since  $N(d)_j = N_{j+d}$ .

We write

$\mathrm{GrMod} A :=$  category of graded right  $A$ -modules with morphism spaces  $\mathrm{Hom}_A^{\mathbb{Z}}(M, N)$ ,  
 $\mathrm{grmod} A :=$  full subcategory of  $\mathrm{GrMod} A$  consisting of locally finite modules.

Forgetting the grading gives a functor  $\text{GrMod } A \rightarrow \text{Mod } A$ . The distinction between its source and target is the grading on objects and the degree condition on morphisms.

For  $d \in \mathbb{Z}$ , the shift of Definition 1.20 is a functor

$$(d) : \text{GrMod } A \rightarrow \text{GrMod } A.$$

It sends  $f : M \rightarrow N$  to the same underlying map  $f(d) : M(d) \rightarrow N(d)$ . Since  $M(d)(-d) = M$  and  $f(d)(-d) = f$ , the shift is an equivalence with inverse  $(-d)$ ; it restricts to an equivalence  $\text{grmod } A \rightarrow \text{grmod } A$ .

**Lemma 1.25.** *For graded right  $A$ -modules  $M, N$ , the following inclusion is an equality when  $M$  is finitely generated:*

$$\bigoplus_{d \in \mathbb{Z}} \text{Hom}_A^{\mathbb{Z}}(M, N(d)) \subset \text{Hom}_A(M, N).$$

**Proof** For  $f \in \text{Hom}_A(M, N)$ , define  $f_d(m) = f(m)_{j+d}$  when  $m \in M_j$ , and extend linearly. For  $a \in A_i$ , we have  $f_d(ma) = f(ma)_{j+i+d} = f(m)_{j+d}a$ , so  $f_d$  is  $A$ -linear of degree  $d$ . The sum of the homogeneous Hom spaces is direct: on  $M_j$ , distinct degrees have values in distinct summands of  $N$ . If  $M$  is finitely generated, the homogeneous components of a finite generating set form a finite homogeneous generating set. Only finitely many  $f_d$  are non-zero on these homogeneous generators, hence on  $M$ , giving  $f = \sum_d f_d$ .  $\square$

## 1.6 Homogeneous ideals and quotients

**Definition 1.26.** Let  $A = \bigoplus_{i \in \mathbb{Z}} A_i$  be a graded algebra. A right ideal  $I \subset A$  is *homogeneous* if

$$I = \bigoplus_{i \in \mathbb{Z}} (I \cap A_i).$$

Equivalently, whenever  $a \in I$  and  $a = a_0 + \cdots + a_N$  is its decomposition into homogeneous components, each  $a_i$  also belongs to  $I$ .

For quotient algebras we need two-sided ideals. If  $I$  is a homogeneous two-sided ideal, then

$$A/I = \bigoplus_{i \in \mathbb{Z}} A_i / (I \cap A_i)$$

is again a graded algebra.

**Example 1.27.** In  $A = \mathbb{k}A_3$  with the grading of Example 1.13, the two-sided ideal  $I = (e_{13}) = \mathbb{k}e_{13}$  is homogeneous. Writing  $\bar{e}_{ij}$  for the image of  $e_{ij}$ ,

$$(A/I)_0 \cong \mathbb{k}^{\oplus 3}, \quad (A/I)_1 = \mathbb{k}\bar{e}_{12} \oplus \mathbb{k}\bar{e}_{23}, \quad (A/I)_d = 0 \quad (d \geq 2).$$

In particular,  $\bar{e}_{12}\bar{e}_{23} = 0$ .

**Example 1.28.** Let  $A = \mathbb{k}[x]$  with  $\deg x = 1$ , and let  $I = (x - x^2)$ . The ideal  $I$  is two-sided because  $A$  is commutative. Every element of  $I$  vanishes at  $x = 1$ , whereas neither  $x$  nor  $x^2$  does. Thus  $x - x^2 \in I$  but  $x, x^2 \notin I$ , so  $I$  is not homogeneous. Writing  $\bar{f} = f + I$ , we have  $0 \neq \bar{x} = \overline{x^2}$ . The images of  $A_1$  and  $A_2$  therefore intersect non-trivially, so  $A/I$  does not inherit the quotient grading. As an ungraded algebra, the quotient is semisimple:

$$\mathbb{k}[x]/(x - x^2) \xrightarrow{\sim} \mathbb{k} \oplus \mathbb{k}, \quad \bar{f} \mapsto (f(0), f(1)).$$

The inverse sends  $(a, b)$  to  $a + (b - a)\bar{x}$ .

**Definition 1.29.** The *positive degree ideal* of a graded algebra  $A$  is  $A_{>0} := \bigoplus_{i>0} A_i$ .<sup>1</sup>

The subspace  $A_{>0}$  is a homogeneous two-sided ideal. The quotient map

$$\pi : A \longrightarrow A/A_{>0} \cong A_0, \quad \sum_{i \in \mathbb{Z}} a_i \mapsto a_0,$$

makes  $A_0$  a graded  $A$ -bimodule concentrated in degree 0, with  $a \cdot s \cdot b = \pi(a)s\pi(b)$  for  $a, b \in A$  and  $s \in A_0$ .

## 1.7 Tensor products of vector spaces

**Definition 1.30.** For vector spaces  $U, V$ , the *tensor product*  $U \otimes_{\mathbb{k}} V$  is equipped with a bilinear map  $(u, v) \mapsto u \otimes v$  such that every bilinear map  $b : U \times V \rightarrow W$  has a unique linear factorisation:

$$\begin{array}{ccc} U \times V & \xrightarrow{(u,v) \mapsto u \otimes v} & U \otimes_{\mathbb{k}} V \\ & \searrow b & \downarrow \exists! \tilde{b} \\ & & W. \end{array}$$

Choose bases  $(u_i)_{i \in I}$  of  $U$  and  $(v_j)_{j \in J}$  of  $V$ . Construct  $U \otimes_{\mathbb{k}} V$  as the vector space with basis  $\{u_i \otimes v_j \mid i \in I, j \in J\}$ , and extend  $(u_i, v_j) \mapsto u_i \otimes v_j$  bilinearly. For  $b : U \times V \rightarrow W$  bilinear, the unique required map is

$$\tilde{b} : U \otimes_{\mathbb{k}} V \longrightarrow W, \quad u_i \otimes v_j \longmapsto b(u_i, v_j).$$

Let  $T, T'$  be vector spaces with bilinear maps  $t : U \times V \rightarrow T$  and  $t' : U \times V \rightarrow T'$  satisfying the universal property in Definition 1.30. The universal properties give unique linear maps  $a : T \rightarrow T'$  and  $b : T' \rightarrow T$  with  $a \circ t = t'$  and  $b \circ t' = t$ . The universal property of  $(T, t)$ , applied to the bilinear map  $t : U \times V \rightarrow T$ , gives a unique linear map  $h : T \rightarrow T$  with  $h \circ t = t$ . Both  $b \circ a$  and  $\text{id}_T$  have this property, since

$$(b \circ a) \circ t = b \circ t' = t = \text{id}_T \circ t.$$

Hence  $b \circ a = \text{id}_T$ . Likewise,  $(a \circ b) \circ t' = a \circ t = t' = \text{id}_{T'} \circ t'$ , so uniqueness for  $(T', t')$  gives  $a \circ b = \text{id}_{T'}$ . Thus the tensor product is independent of the chosen bases up to the unique isomorphism carrying  $t(u, v)$  to  $t'(u, v)$ .

For finite bases with  $|I| = r$  and  $|J| = s$ , we obtain  $\dim_{\mathbb{k}}(U \otimes_{\mathbb{k}} V) = rs$ . We use the canonical identifications

$$\begin{aligned} (U \otimes_{\mathbb{k}} V) \otimes_{\mathbb{k}} W &\xrightarrow{\sim} U \otimes_{\mathbb{k}} (V \otimes_{\mathbb{k}} W), & (u \otimes v) \otimes w &\longmapsto u \otimes (v \otimes w), \\ \mathbb{k} \otimes_{\mathbb{k}} U &\xrightarrow{\sim} U, & c \otimes u &\longmapsto cu, \end{aligned}$$

and set  $U^{\otimes_{\mathbb{k}} 0} := \mathbb{k}$ .

## 1.8 Enveloping algebras and bimodules

**Definition 1.31.** For a  $\mathbb{k}$ -algebra  $A$ , its *enveloping algebra* is  $A^{\text{en}} := A \otimes_{\mathbb{k}} A^{\text{op}}$ , with multiplication extended bilinearly from

$$(a \otimes b^{\text{op}})(a' \otimes (b')^{\text{op}}) = aa' \otimes (b'b)^{\text{op}}, \quad 1_{A^{\text{en}}} = 1_A \otimes 1_A^{\text{op}} \quad (a, b, a', b' \in A).$$

---

<sup>1</sup>It is also called the augmentation ideal; we avoid that terminology.

An  $A$ -bimodule becomes a right  $A^{\text{en}}$ -module by

$$m \cdot (a \otimes b^{\text{op}}) = bma \quad (a, b \in A, m \in M). \quad (1.3)$$

The first tensor factor records the right action, and the second the left action:

$$(m \cdot (a \otimes b^{\text{op}})) \cdot (a' \otimes (b')^{\text{op}}) = b'bma a' = m \cdot (aa' \otimes (b'b)^{\text{op}}).$$

Conversely, for  $M \in \text{Mod } A^{\text{en}}$ , define  $ma = m \cdot (a \otimes 1)$  and  $bm = m \cdot (1 \otimes b^{\text{op}})$ . The two actions commute because  $a \otimes 1$  and  $1 \otimes b^{\text{op}}$  commute in  $A^{\text{en}}$ . For  $A$ -bimodules  $M, N$ , a linear map  $u : M \rightarrow N$  satisfies

$$u(bma) = bu(m)a \quad (a, b \in A, m \in M) \iff u \in \text{Hom}_{A^{\text{en}}}(M, N).$$

Thus  $\text{Mod } A^{\text{en}}$  is also the category of  $A$ -bimodules and maps preserving both actions.

*Remark 1.32.* For a finite-dimensional semisimple  $\mathbb{k}$ -algebra  $S$ , semisimplicity of  $S^{\text{en}}$  requires an additional condition. For a field extension  $K/\mathbb{k}$ , give  $K \otimes_{\mathbb{k}} S$  the multiplication  $(c \otimes s)(d \otimes t) = cd \otimes st$ . The algebra  $S$  is called *separable over  $\mathbb{k}$*  if  $K \otimes_{\mathbb{k}} S$  is semisimple for every field extension  $K/\mathbb{k}$ . We record the following fact without proof:

$$S^{\text{en}} \text{ is semisimple} \iff S \text{ is separable over } \mathbb{k}.$$

For a product of matrix algebras, every field extension  $K/\mathbb{k}$  gives

$$K \otimes_{\mathbb{k}} \prod_{j=1}^r \text{Mat}_{n_j}(\mathbb{k}) \xrightarrow{\sim} \prod_{j=1}^r \text{Mat}_{n_j}(K), \quad c \otimes (a_j)_j \mapsto (ca_j)_j.$$

The target is semisimple by Theorem 1.9. Thus products of matrix algebras over  $\mathbb{k}$  are separable; in particular, every finite-dimensional semisimple algebra over an algebraically closed field is separable. Semisimplicity of  $S$  alone does not imply semisimplicity of  $S^{\text{en}}$  over an arbitrary field.

**Example 1.33.** Let  $\mathbb{k} = \mathbb{F}_2(t)$ , where  $t$  is an indeterminate, and let  $S = \mathbb{k}[u]/(u^2 - t)$ , also writing  $u$  for its residue class. The element  $t$  is not a square in  $\mathbb{k}$ : an equality  $p(t)^2 = tq(t)^2$  for non-zero polynomials would give  $2 \deg p = 1 + 2 \deg q$ . Thus  $u^2 - t$  is irreducible and  $S$  is a field, hence semisimple. However, since  $S$  is commutative and has characteristic 2,

$$S^{\text{en}} = S \otimes_{\mathbb{k}} S \cong S[v]/(v^2 - t) = S[v]/((v - u)^2) \cong S[\varepsilon]/(\varepsilon^2).$$

Here the first tensor factor supplies the coefficients,  $v$  corresponds to  $1 \otimes u$ , and  $\varepsilon = 1 \otimes u - u \otimes 1$ . The ideal  $S\varepsilon$  is non-zero, has square zero, and is the unique maximal ideal, so  $\text{rad}(S^{\text{en}}) = S\varepsilon \neq 0$ . Thus  $S^{\text{en}}$  is not semisimple, and the field extension  $S/\mathbb{k}$  already shows that  $S$  is not separable over  $\mathbb{k}$ .

## 1.9 Direct sums and direct products

For a family  $(M_i)_{i \in I}$  of right  $A$ -modules, the *direct product* and *direct sum* are

$$\begin{aligned} \prod_{i \in I} M_i &= \{(m_i)_{i \in I} \mid m_i \in M_i \text{ for all } i \in I\}, \\ \bigoplus_{i \in I} M_i &= \{(m_i)_{i \in I} \in \prod_{i \in I} M_i \mid m_i = 0 \text{ for all but finitely many } i\}. \end{aligned}$$

Addition and the  $A$ -action are componentwise; the same definitions apply to vector spaces by taking  $A = \mathbb{k}$ . Thus  $\bigoplus_i M_i \subset \prod_i M_i$ , with equality for finite  $I$ . For example,  $(1, 1, \dots) \in \prod_{i \geq 0} \mathbb{k}$  does not belong to  $\bigoplus_{i \geq 0} \mathbb{k}$ . In a direct sum, an element is a finite sum of its components; no notion of convergence is involved.

## 1.10 Graded $\mathbb{k}$ -linear duals

A linear functional on a graded vector space need not have a degree: it may be non-zero on infinitely many homogeneous components. To obtain a graded vector space, we retain only finite sums of homogeneous functionals.

**Definition 1.34.** For a graded vector space  $V = \bigoplus_{i \in \mathbb{Z}} V_i$ , its *graded  $\mathbb{k}$ -linear dual* is the graded vector space defined using the linear dual  $D$  of Example 1.7 by

$$\mathbb{D}(V) = \bigoplus_{i \in \mathbb{Z}} \mathbb{D}(V)_i \quad \text{with} \quad \mathbb{D}(V)_i = D(V_{-i}).$$

For general infinite-dimensional  $V$ , we have  $\mathbb{D}(V) \subset D(V)$ , whereas restriction to the homogeneous components gives  $D(V) \cong \prod_{i \in \mathbb{Z}} D(V_i)$ . An arbitrary family of functionals on the  $V_i$  defines a functional on  $V$ , since each vector has only finitely many non-zero components. These spaces coincide when  $V$  is finite-dimensional.

Linear duality reverses the side of a module action. For a right  $A$ -module  $M$  and a left  $A$ -module  $N$ , define

$$(af)(m) := f(ma), \quad (ga)(n) := g(an) \quad (a \in A, f \in D(M), g \in D(N)). \quad (1.4)$$

Here  $m \in M$  and  $n \in N$ ; the first formula makes  $D(M)$  a left module and the second makes  $D(N)$  a right module. For graded modules, the same formulas preserve the graded  $\mathbb{k}$ -linear dual: if  $a \in A_p$  and  $f \in \mathbb{D}(M)_q$ , then  $af$  vanishes outside  $M_{-p-q}$  and hence has degree  $p + q$ .

For a module homomorphism  $u : M \rightarrow N$ , precomposition defines  $D(u) : D(N) \rightarrow D(M)$  by  $D(u)(f) = f \circ u$ . If  $u$  preserves degrees, this restricts to  $\mathbb{D}(u) : \mathbb{D}(N) \rightarrow \mathbb{D}(M)$ , also of degree 0. Thus we obtain functors

$$D : (\text{Mod } A)^{\text{op}} \longrightarrow \text{Mod}(A^{\text{op}}), \quad \mathbb{D} : (\text{GrMod } A)^{\text{op}} \longrightarrow \text{GrMod}(A^{\text{op}}). \quad (1.5)$$

Indeed,  $D(\text{id}_M) = \text{id}_{D(M)}$  and  $D(v \circ u) = D(u) \circ D(v)$  for composable  $u, v$ , and the same identities hold for  $\mathbb{D}$ . Replacing  $A$  by  $A^{\text{op}}$  gives the corresponding functors from left modules to right modules.

**Example 1.35.** Let  $A = \mathbb{k}A_3$  and  $P_1 = e_{11}A$  as in Example 1.21. The functionals  $\varepsilon_{1j}(e_{1\ell}) = \delta_{j\ell}$  for  $1 \leq j, \ell \leq 3$  form a basis of  $\mathbb{D}(P_1)$ , with

$$\deg \varepsilon_{1j} = 1 - j, \quad e_{12}\varepsilon_{12} = \varepsilon_{11}, \quad e_{23}\varepsilon_{13} = \varepsilon_{12}, \quad e_{jj}\varepsilon_{1j} = \varepsilon_{1j}.$$

The actions follow from (1.4).

**Definition 1.36.** Let  $L, M, N \in \text{Mod } A$ , with  $i : L \rightarrow M$  and  $p : M \rightarrow N$  module homomorphisms. The sequence  $L \xrightarrow{i} M \xrightarrow{p} N$  is *exact at  $M$*  if

$$\text{Im } i = \text{Ker } p, \quad \text{that is,} \quad \{i(\ell) \mid \ell \in L\} = \{m \in M \mid p(m) = 0\}.$$

A sequence of module homomorphisms is *exact* if it is exact at every term with an incoming and an outgoing map. A *short exact sequence* is an exact sequence

$$0 \longrightarrow L \xrightarrow{i} M \xrightarrow{p} N \longrightarrow 0, \quad \text{Ker } i = 0, \quad \text{Im } i = \text{Ker } p, \quad \text{Im } p = N. \quad (1.6)$$

A sequence in  $\text{GrMod } A$  has maps of degree 0 and is exact if the underlying sequence in  $\text{Mod } A$  is exact.

In (1.6), the map  $i$  identifies  $L$  with the submodule  $\text{Ker } p \subset M$ , and  $p$  gives  $M/i(L) \cong N$ . For degree 0 maps  $i, p$ , exactness is equivalent to exactness of

$$0 \longrightarrow L_j \xrightarrow{i|_{L_j}} M_j \xrightarrow{p|_{M_j}} N_j \longrightarrow 0 \quad \text{for every } j \in \mathbb{Z}.$$

Indeed,  $i$  and  $p$  preserve the direct sum decompositions, so their kernels and images decompose into the kernels and images of the restrictions.

**Example 1.37.** Let  $A = \mathbb{k}A_2$ . The inclusion and quotient maps give a short exact sequence in  $\text{GrMod } A$ :

$$0 \longrightarrow \mathbb{k}e_{12} \xrightarrow{i} e_{11}A \xrightarrow{p} e_{11}A/\mathbb{k}e_{12} \longrightarrow 0, \quad i(ce_{12}) = ce_{12}, \quad p(m) = \overline{m}.$$

The source is concentrated in degree 1, the quotient in degree 0, and  $\text{Ker } p = \mathbb{k}e_{12} = \text{Im } i$ .

**Lemma 1.38.** The functor  $\mathbb{D}$  in (1.5) restricts to a duality  $(\text{grmod } A)^{\text{op}} \rightarrow \text{grmod}(A^{\text{op}})$ . The coevaluation map and its inverse are natural isomorphisms

$$\begin{aligned} \text{coev}_M : M &\xrightarrow{\sim} \mathbb{D}(\mathbb{D}(M)), & m &\longmapsto (f \mapsto f(m)), \\ \text{ev}_M := \text{coev}_M^{-1} : \mathbb{D}(\mathbb{D}(M)) &\xrightarrow{\sim} M & (M \in \text{grmod } A). \end{aligned} \tag{1.7}$$

Thus  $\text{ev}_M(\Phi)$  is the unique  $m \in M$  such that  $\Phi(f) = f(m)$  for every  $f \in \mathbb{D}(M)$ . Moreover,  $\mathbb{D}$  takes every short exact sequence  $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$  to a short exact sequence  $0 \rightarrow \mathbb{D}(N) \rightarrow \mathbb{D}(M) \rightarrow \mathbb{D}(L) \rightarrow 0$ .

**Proof** The map  $\text{coev}_M$  in (1.7) is an isomorphism in each degree by Example 1.7, and (1.4) makes it  $A$ -linear. For  $u : M \rightarrow N$  of degree 0,  $m \in M$ , and  $f \in \mathbb{D}(N)$ ,

$$(\mathbb{D}(\mathbb{D}(u))(\text{coev}_M(m)))(f) = \text{coev}_M(m)(f \circ u) = f(u(m)) = \text{coev}_N(u(m))(f).$$

Hence  $\mathbb{D}(\mathbb{D}(u)) \circ \text{coev}_M = \text{coev}_N \circ u$ , which is naturality as in (1.1). Taking inverses of the coevaluation maps gives  $\text{ev}_N \circ \mathbb{D}(\mathbb{D}(u)) = u \circ \text{ev}_M$ . Replacing  $A$  by  $A^{\text{op}}$  gives the other natural isomorphism required for a duality. For a short exact sequence as in (1.6), the maps on the dual sequence are  $\mathbb{D}(p) : \mathbb{D}(N) \rightarrow \mathbb{D}(M)$  and  $\mathbb{D}(i) : \mathbb{D}(M) \rightarrow \mathbb{D}(L)$ . In each degree,  $\mathbb{D}(p)$  is injective because  $p$  is surjective, and  $\mathbb{D}(i)$  is surjective because a functional on a subspace extends to the whole vector space. For  $\lambda \in \mathbb{D}(M)_d$  with  $\lambda \circ i = 0$ , define  $\mu : N \rightarrow \mathbb{k}$  by  $\mu(p(m)) = \lambda(m)$ . This is well-defined because  $\lambda$  vanishes on  $\text{Ker } p = i(L)$ ; since  $p$  preserves degrees,  $\mu \in \mathbb{D}(N)_d$ . Thus  $\lambda = \mathbb{D}(p)(\mu)$ . Conversely,  $p \circ i = 0$  gives  $\mathbb{D}(i) \circ \mathbb{D}(p) = 0$ , proving  $\text{Ker } \mathbb{D}(i) = \text{Im } \mathbb{D}(p)$ .  $\square$

The same argument without grading gives a duality between finite-dimensional right and left  $A$ -modules. If  $A$  is finite-dimensional, these are precisely the finitely generated modules, so  $D : (\text{mod } A)^{\text{op}} \rightarrow \text{mod}(A^{\text{op}})$  is a duality. Without finite dimensionality for  $D$ , or local finiteness for  $\mathbb{D}$ , coevaluation need not be surjective, so its inverse  $\text{ev}$  need not exist.

### 1.11 Left, right, and two-sided duals over $S$

Fix a finite-dimensional semisimple  $\mathbb{k}$ -algebra  $S$ , which is regarded as concentrated in degree 0 when treated as a graded algebra. Besides the  $\mathbb{k}$ -linear dual, we will also work with duals defined by left  $S$ -linear maps and right  $S$ -linear maps. Their definitions differ, even though the resulting spaces can be identified after a choice of form.

**Definition 1.39.** For a right  $S$ -module  $M$  and a left  $S$ -module  $N$ , define their  $S$ -duals by

$$\begin{aligned} M^* &:= \text{Hom}_S(M, S), & (sf)(m) &:= sf(m), \\ {}^*N &:= \text{Hom}_{S^{\text{op}}}(N, S), & (gs)(n) &:= g(n)s. \end{aligned}$$

Here  $s \in S$ ,  $m \in M$ ,  $n \in N$ ,  $f \in M^*$ , and  $g \in {}^*N$ .

$$M \in \text{Mod } S \implies M^* \in \text{Mod}(S^{\text{op}}), \quad N \in \text{Mod}(S^{\text{op}}) \implies {}^*N \in \text{Mod } S.$$

The same implications hold with  $\text{Mod}$  replaced by  $\text{mod}$ , since  $S$  is finite-dimensional.

For a right module map  $u : M \rightarrow N$ , define  $u^* : N^* \rightarrow M^*$  by  $u^*(f) = f \circ u$ ; for a left module map  $u : M \rightarrow N$ , the same recipe defines  ${}^*u : {}^*N \rightarrow {}^*M$ . Precomposition gives functors

$$(-)^* : (\text{Mod } S)^{\text{op}} \longrightarrow \text{Mod}(S^{\text{op}}), \quad {}^*(-) : (\text{Mod}(S^{\text{op}}))^{\text{op}} \longrightarrow \text{Mod } S.$$

Recall from Definition 1.31 and (1.3) that

$$S^{\text{en}} = S \otimes_{\mathbb{k}} S^{\text{op}}, \quad m \cdot (s \otimes t^{\text{op}}) = tms,$$

so  $S$ -bimodules are the objects of  $\text{Mod } S^{\text{en}}$ . All three spaces  $D(M)$ ,  $M^*$ , and  ${}^*M$  inherit bimodule structures:

$$(s\lambda t)(m) = \lambda(tms), \quad (sft)(m) = sf(tm), \quad (sgt)(m) = g(ms)t, \quad (1.8)$$

where  $\lambda \in D(M)$ ,  $f \in M^*$ ,  $g \in {}^*M$ ,  $s, t \in S$ , and  $m \in M$ .

**Definition 1.40.** The *two-sided dual* of an  $S$ -bimodule  $M$  is

$$M^{*\text{en}} := \text{Hom}_{S^{\text{en}}}(M, S^{\text{en}}), \quad (sFt)(m) := (s \otimes t^{\text{op}})F(m) \quad (s, t \in S, F \in M^{*\text{en}}, m \in M).$$

In Definition 1.40,  $M$  has the right action (1.3), so  $F(tms) = F(m)(s \otimes t^{\text{op}})$ . The output is a left  $S^{\text{en}}$ -module by multiplication on the values, which gives the displayed bimodule action. The two-sided dual takes values in  $S^{\text{en}}$ , not in  $S$ . On bimodule morphisms it acts by precomposition, as do  $D$ ,  $(-)^*$ , and  ${}^*(-)$ ; each is a contravariant functor on the category of  $S$ -bimodules. The two-sided dual records both module actions in the target algebra  $S^{\text{en}}$ .

## 1.12 Comparing the duals

To identify  $S$ -dual with  $\mathbb{k}$ -dual, we need a suitable linear functional  $S \rightarrow \mathbb{k}$ .

**Definition 1.41.** A *symmetrising form* on a finite-dimensional  $\mathbb{k}$ -algebra  $S$  is a linear map  $\tau : S \rightarrow \mathbb{k}$  satisfying

$$\tau(st) = \tau(ts) \quad \text{for all } s, t \in S, \quad (\tau(st) = 0 \text{ for all } t \in S) \implies s = 0.$$

An algebra admitting such a form is *symmetric*.

The second condition in Definition 1.41 says that the bilinear form  $(s, t) \mapsto \tau(st)$  is *non-degenerate*: every non-zero  $s \in S$  pairs non-trivially with some  $t \in S$ . The first condition makes this form symmetric, so the analogous assertion holds with the two arguments interchanged.

We will only use the following explicit symmetrising forms.

**Example 1.42.** The following maps are symmetrising forms:

$$\begin{aligned}\tau : \mathbb{k}^{\oplus r} &\longrightarrow \mathbb{k}, & (a_1, \dots, a_r) &\longmapsto \sum_{j=1}^r a_j, \\ \tau : \text{Mat}_n(\mathbb{k}) &\longrightarrow \mathbb{k}, & a &\longmapsto \text{Tr}(a) := \sum_{i=1}^n a_{ii}, \\ \tau : \prod_{j=1}^r \text{Mat}_{n_j}(\mathbb{k}) &\longrightarrow \mathbb{k}, & (a_1, \dots, a_r) &\longmapsto \sum_{j=1}^r \text{Tr}(a_j).\end{aligned}$$

For matrices  $a, b \in \text{Mat}_n(\mathbb{k})$ ,

$$\text{Tr}(ab) = \sum_{i,j=1}^n a_{ij}b_{ji} = \text{Tr}(ba), \quad \text{Tr}(ae_{ji}) = a_{ij} \quad (1 \leq i, j \leq n).$$

Thus  $\text{Tr}(ab) = 0$  for all  $b$  forces  $a = 0$ . For the product, take  $b$  supported on a single matrix factor; the same calculation detects every entry of every  $a_j$ . The first form is the case  $n_j = 1$  for all  $j$ .

**Proposition 1.43 (Dual comparison).** Choose a symmetrising form  $\tau$  on  $S$ . For finite-dimensional right and left  $S$ -modules  $M$ , respectively, the maps

$$\alpha_M : M^* \longrightarrow D(M), \quad f \longmapsto \tau \circ f, \quad \beta_M : {}^*M \longrightarrow D(M), \quad g \longmapsto \tau \circ g \quad (1.9)$$

are isomorphisms of left and right  $S$ -modules, respectively. For fixed  $\tau$ , they are natural isomorphisms of the corresponding contravariant functors. For example, every right  $S$ -module map  $u : M \rightarrow N$  between finite-dimensional modules gives a commuting square

$$\begin{array}{ccc} N^* & \xrightarrow{\alpha_N} & D(N) \\ u^* \downarrow & & \downarrow D(u) \\ M^* & \xrightarrow{\alpha_M} & D(M). \end{array} \quad (1.10)$$

The corresponding squares for  $\beta$  commute for left  $S$ -module maps.

If  $M$  is an  $S$ -bimodule,  $\alpha_M$  and  $\beta_M$  preserve both actions from (1.8). There is also an isomorphism of  $S$ -bimodules  $\gamma_M : M^{\text{en}} \rightarrow M^*$  given by

$$\gamma_M(F)(m) = \sum_j a_j \tau(b_j) \quad \text{where} \quad F(m) = \sum_j a_j \otimes b_j^{\text{op}}. \quad (1.11)$$

For fixed  $\tau$ ,  $\gamma$  is a natural isomorphism of contravariant functors on finite-dimensional  $S$ -bimodules.

**Proof** Non-degeneracy makes  $S \rightarrow D(S)$ ,  $a \mapsto (s \mapsto \tau(as))$ , injective, hence bijective because both spaces have the same finite dimension. Thus for  $\lambda \in D(M)$ , according to whether  $M$  is a right or left module, there exist unique  $f(m)$  or  $g(m)$  in  $S$  satisfying

$$\tau(f(m)s) = \lambda(ms), \quad \tau(g(m)s) = \lambda(sm) \quad (s \in S, m \in M). \quad (1.12)$$

The inverse of  $S \rightarrow D(S)$  is linear, so  $m \mapsto f(m)$  and  $m \mapsto g(m)$  are linear. By non-degeneracy, (1.12) gives  $f(ms) = f(m)s$  in the right module case and  $g(sm) = sg(m)$  in the left module case. Taking  $s = 1$  gives  $\alpha_M(f) = \lambda$  or  $\beta_M(g) = \lambda$ , respectively. Conversely, right or left  $S$ -linearity forces any preimage of  $\lambda$  to satisfy the corresponding identity in (1.12). The uniqueness in (1.12) therefore proves that  $\alpha_M$  and  $\beta_M$  are bijective. For  $s \in S$ ,  $f \in M^*$ , and  $m \in M$ , symmetry and right  $S$ -linearity give

$$\alpha_M(sf)(m) = \tau(sf(m)) = \tau(f(m)s) = \tau(f(ms)) = (s\alpha_M(f))(m).$$

Likewise,  $\beta_M(gs)(m) = \tau(g(m)s) = \tau(g(sm)) = \tau(g(sm)) = (\beta_M(g)s)(m)$  for a left module  $M$  and  $g \in {}^*M$ . For (1.10), both routes send  $f \in N^*$  to  $\tau \circ f \circ u$ ; the calculation for  $\beta$  is identical.

For bimodule compatibility, use (1.8) and

$$\tau(sf(tm)) = \tau(f(tm)s) = \tau(f(tms)), \quad \tau(g(ms)t) = \tau(tg(ms)) = \tau(g(tms)).$$

The form  $\tau_{\text{en}}(a \otimes b^{\text{op}}) = \tau(a)\tau(b)$  is symmetrising on  $S^{\text{en}}$ . Indeed,  $\tau_{\text{en}}((a \otimes b^{\text{op}})(c \otimes d^{\text{op}})) = \tau(ac)\tau(db) = \tau(ca)\tau(bd)$ , and its pairing is the tensor product of two non-degenerate pairings. The formula (1.11) satisfies

$$\gamma_M(F)(ms) = \gamma_M(F)(m)s, \quad \gamma_M(sFt) = s\gamma_M(F)t, \quad \alpha_M(\gamma_M(F)) = \tau_{\text{en}} \circ F.$$

The argument (1.12), applied to  $S^{\text{en}}$  and  $\tau_{\text{en}}$ , makes  $F \mapsto \tau_{\text{en}} \circ F$  an isomorphism. Thus  $\gamma_M$  is an isomorphism. For a bimodule map  $u : M \rightarrow N$  and  $F \in N^{*\text{en}}$ , the formula (1.11) gives  $\gamma_M(F \circ u) = \gamma_N(F) \circ u$ , proving naturality.

□

*Remark 1.44.* The identifications (1.9) depend on  $\tau$ . For instance,  $\tau(a_1, \dots, a_r) = \sum_i c_i a_i$  is symmetrising on  $\mathbb{k}^{\oplus r}$  for any  $c_i \neq 0$ .

The two-sided comparison (1.11) also depends on  $\tau$ ; it does not require  $S^{\text{en}}$  to be semisimple.

**Definition 1.45.** For graded right and left  $S$ -modules  $M$ , respectively, define their *graded  $S$ -duals* by

$$M^{(*)} := \bigoplus_{i \in \mathbb{Z}} (M_{-i})^*, \quad ({}^*)M := \bigoplus_{i \in \mathbb{Z}} {}^*(M_{-i}),$$

where each indicated summand has degree  $i$ .

For a degree 0 right  $S$ -module map  $u : M \rightarrow N$ , define  $u^{(*)} : N^{(*)} \rightarrow M^{(*)}$  by

$$(u^{(*)})_i : (N_{-i})^* \longrightarrow (M_{-i})^*, \quad f \longmapsto f \circ u|_{M_{-i}} \quad (i \in \mathbb{Z}).$$

For a degree 0 left module map  $u : M \rightarrow N$ , the same formula defines  $({}^*)u : ({}^*)N \rightarrow ({}^*)M$ . With the actions from Definition 1.39, we obtain functors

$$\begin{aligned} (-)^{(*)} : (\text{GrMod } S)^{\text{op}} &\longrightarrow \text{GrMod}(S^{\text{op}}), \\ ({}^*)(-) : (\text{GrMod}(S^{\text{op}}))^{\text{op}} &\longrightarrow \text{GrMod } S. \end{aligned} \tag{1.13}$$

Each functor in (1.13) restricts to locally finite modules: replace  $\text{GrMod}$  by  $\text{grmod}$  in its source and target. Indeed, for  $R = S$  or  $S^{\text{op}}$  and  $M \in \text{grmod } R$ ,

$$\dim_{\mathbb{k}} \text{Hom}_R(M_{-i}, R) \leq (\dim_{\mathbb{k}} M_{-i})(\dim_{\mathbb{k}} R) < \infty \quad (i \in \mathbb{Z}). \tag{1.14}$$

Parentheses on the stars distinguish the graded constructions from the full  $S$ -duals. In the notation (1.2),

$$M^{(*)} = \bigoplus_{i \in \mathbb{Z}} \text{Hom}_S^{\mathbb{Z}}(M, S(i)), \quad ({}^*)M = \bigoplus_{i \in \mathbb{Z}} \text{Hom}_{S^{\text{op}}}^{\mathbb{Z}}(M, S(i)).$$

For a graded  $S$ -bimodule  $M$ , define its *graded two-sided dual* by

$$M^{(*\text{en})} := \bigoplus_{i \in \mathbb{Z}} (M_{-i})^{*\text{en}}, \quad (M^{(*\text{en})})_i = (M_{-i})^{*\text{en}}.$$

The bimodule actions in (1.8) and Definition 1.40 give functors

$$(-)^{(*)}, {}^{(*)}(-), (-)^{(*\text{en})} : (\text{GrMod } S^{\text{en}})^{\text{op}} \longrightarrow \text{GrMod } S^{\text{en}}. \quad (1.15)$$

On a degree 0 bimodule map  $u : M \rightarrow N$ , each functor acts by precomposition in each degree; in particular,

$$(u^{(*\text{en})})_i : (N_{-i})^{*\text{en}} \longrightarrow (M_{-i})^{*\text{en}}, \quad F \longmapsto F \circ u|_{M_{-i}}.$$

These functors restrict to locally finite bimodules: the dimension bound (1.14) also applies with  $R = S^{\text{en}}$ .

**Corollary 1.46.** *Fix a symmetrising form  $\tau : S \rightarrow \mathbb{k}$ . There are natural isomorphisms*

$$\begin{aligned} \alpha : (-)^{(*)} &\xrightarrow{\sim} \mathbb{D} \quad \text{as functors } (\text{grmod } S)^{\text{op}} \rightarrow \text{grmod}(S^{\text{op}}), \\ \beta : {}^{(*)}(-) &\xrightarrow{\sim} \mathbb{D} \quad \text{as functors } (\text{grmod}(S^{\text{op}}))^{\text{op}} \rightarrow \text{grmod } S. \end{aligned}$$

*On locally finite graded  $S$ -bimodules, there are natural isomorphisms of functors  $(\text{grmod } S^{\text{en}})^{\text{op}} \rightarrow \text{grmod } S^{\text{en}}$ :*

$$(-)^{(*\text{en})} \xrightarrow[\gamma]{\sim} (-)^{(*)} \xrightarrow[\alpha]{\sim} \mathbb{D} \xleftarrow[\beta]{\sim} {}^{(*)}(-).$$

**Proof** For a locally finite graded module  $M$ ,  $\alpha_M$  and  $\beta_M$  are the direct sums over  $i \in \mathbb{Z}$  of the corresponding maps on  $M_{-i}$  in Proposition 1.43. They are isomorphisms of degree 0, and their naturality squares are the direct sums of the squares in (1.10) and its counterpart for  $\beta$ .

For a locally finite graded bimodule  $M$ , take the direct sum of the maps  $\gamma_{M_{-i}}$  from (1.11). The same degree-wise argument proves that  $\gamma_M$  is an isomorphism of graded bimodules and is natural in  $M$ .

□

For a graded algebra  $A$  with  $S = A_0$ , Corollary 1.46 transports the  $A$ -action (1.4) to the graded  $S$ -duals of a locally finite graded  $A$ -module:

$$af = \alpha_M^{-1}(a\alpha_M(f)) \quad \text{for right } M, \quad ga = \beta_M^{-1}(\beta_M(g)a) \quad \text{for left } M.$$

Here  $\alpha_M(f) = \tau \circ f$  and  $\beta_M(g) = \tau \circ g$  are the restrictions of (1.9);  $a \in A$ ,  $f \in M^{(*)}$ , and  $g \in {}^{(*)}M$ . The restrictions to  $S$  agree with Definition 1.39. These remain duals over  $S$ , not duals taking values in  $A$ . The graded  $S$ -duals are defined without  $\tau$ , but these transported  $A$ -actions depend on  $\tau$ . The canonical graded  $A$ -module is  $\mathbb{D}(M)$ .

**Example 1.47.** *Let  $A = \mathbb{k}A_2$ ,  $S = \mathbb{k}e_{11} \oplus \mathbb{k}e_{22}$ , and  $M = e_{11}A$ . For  $c_1, c_2 \in \mathbb{k} \setminus \{0\}$ , choose the symmetrising form*

$$\tau : S \longrightarrow \mathbb{k}, \quad ae_{11} + be_{22} \longmapsto c_1a + c_2b \quad (a, b \in \mathbb{k}). \quad (1.16)$$

*Define right  $S$ -linear maps*

$$\begin{aligned} \varphi_0 : M &\longrightarrow S, \quad ae_{11} + be_{12} \longmapsto ae_{11}, \\ \varphi_1 : M &\longrightarrow S, \quad ae_{11} + be_{12} \longmapsto be_{22} \quad (a, b \in \mathbb{k}). \end{aligned} \quad (1.17)$$

*Their degrees in  $M^{(*)}$  are 0, -1, respectively, and*

$$\alpha_M(e_{12}\varphi_1)(e_{11}) = \alpha_M(\varphi_1)(e_{12}) = c_2, \quad e_{12}\varphi_1 = \frac{c_2}{c_1}\varphi_0.$$

*Thus changing the form can change the action on the same vector space. Two choices give isomorphic  $A$ -modules by composing their identifications with  $\mathbb{D}(M)$ .*

**Example 1.48.** Let  $S = \mathbb{k}^{\oplus r}$  with coordinate idempotents  $e_1, \dots, e_r$  and  $\tau(e_i) = 1$ . Thus  $1 = \sum_i e_i$  and  $e_i e_j = \delta_{ij} e_i$ ; in the diagonal matrix realisation,  $e_i = e_{ii}$ . For an  $S$ -bimodule  $M$ , write its elements in blocks:

$$M \cong \begin{pmatrix} e_1 M e_1 & \cdots & e_1 M e_r \\ \vdots & \ddots & \vdots \\ e_r M e_1 & \cdots & e_r M e_r \end{pmatrix}, \quad m \mapsto (e_i m e_j)_{i,j}.$$

The block matrix denotes the direct sum of its entries; its inverse sends  $(m_{ij})_{i,j}$  to  $\sum_{i,j} m_{ij}$ . The action  $(s\lambda t)(m) = \lambda(tms)$  from (1.8) reverses the blocks:

$$D(M) \cong \begin{pmatrix} D(e_1 M e_1) & \cdots & D(e_r M e_1) \\ \vdots & \ddots & \vdots \\ D(e_1 M e_r) & \cdots & D(e_r M e_r) \end{pmatrix}, \quad e_j D(M) e_i \cong D(e_i M e_j).$$

The last isomorphism restricts a functional to  $e_i M e_j$ ; its inverse extends that functional by zero on the other blocks.

For  $V = (\mathbb{k}A_2)_1 = \mathbb{k}e_{12}$ , the block positions are

$$V = \begin{pmatrix} 0 & \mathbb{k} \\ 0 & 0 \end{pmatrix}, \quad D(V) \cong \begin{pmatrix} 0 & 0 \\ \mathbb{k} & 0 \end{pmatrix}.$$

Let  $\varepsilon \in D(V)$  satisfy  $\varepsilon(e_{12}) = 1$ . Then  $e_2 \varepsilon e_1 = \varepsilon$ , and  $\varepsilon$  has degree  $-1$  in  $\mathbb{D}(V)$ . Under (1.9), the corresponding right and left  $S$ -linear maps are

$$\alpha_V^{-1}(\varepsilon) : V \rightarrow S, \quad e_{12} \mapsto e_2, \quad \beta_V^{-1}(\varepsilon) : V \rightarrow S, \quad e_{12} \mapsto e_1.$$

Thus the same scalar-valued functional corresponds to distinct  $S$ -valued maps on the two sides.

**Lemma 1.49.** Coevaluation gives natural isomorphisms of functors

$$\mathrm{id}_{\mathrm{mod} S} \xrightarrow{\sim} {}^*((-)^*), \quad \mathrm{id}_{\mathrm{mod}(S^{\mathrm{op}})} \xrightarrow{\sim} ({}^*(-))^*,$$

with components

$$\begin{aligned} \mathrm{coev}_M : M &\xrightarrow{\sim} {}^*(M^*), \quad m \mapsto (f \mapsto f(m)) \quad (M \in \mathrm{mod} S), \\ \mathrm{coev}_N : N &\xrightarrow{\sim} ({}^*N)^*, \quad n \mapsto (g \mapsto g(n)) \quad (N \in \mathrm{mod}(S^{\mathrm{op}})). \end{aligned}$$

Thus  $(-)^* : (\mathrm{mod} S)^{\mathrm{op}} \rightarrow \mathrm{mod}(S^{\mathrm{op}})$  and  ${}^*(-) : (\mathrm{mod}(S^{\mathrm{op}}))^{\mathrm{op}} \rightarrow \mathrm{mod} S$  are inverse dualities. For  $S$ -bimodules, the coevaluation maps preserve both actions. No symmetrising form is required.

**Proof** For  $u : M \rightarrow N$  in  $\mathrm{mod} S$ ,  $m \in M$ , and  $f \in N^*$ ,

$$({}^*(u^*)(\mathrm{coev}_M(m)))(f) = (u^*f)(m) = f(u(m)) = \mathrm{coev}_N(u(m))(f).$$

Hence  ${}^*(u^*) \circ \mathrm{coev}_M = \mathrm{coev}_N \circ u$ . For  $M = S$ , coevaluation is an isomorphism: its inverse sends  $F \in {}^*(S^*)$  to  $F(\mathrm{id}_S)$ . Indeed, every  $f \in S^*$  satisfies  $f(s) = f(1)s$ , so  $F(f) = f(1)F(\mathrm{id}_S) = f(F(\mathrm{id}_S))$ . Restriction to summands identifies the dual of a finite direct sum with the direct sum of the duals; coevaluation therefore is an isomorphism for  $S^{\oplus n}$ . By Corollary 1.10,  $M \oplus L \cong S^{\oplus n}$  for some  $L$ . Under the direct sum identifications,  $\mathrm{coev}_{M \oplus L} = \mathrm{coev}_M \oplus \mathrm{coev}_L$ , so  $\mathrm{coev}_M$  is an isomorphism. Replacing  $S$  by  $S^{\mathrm{op}}$  proves the left module statement. For a bimodule  $M$ , (1.8) gives

$$(s \mathrm{coev}_M(m) t)(f) = f(sm)t = f(smt) = \mathrm{coev}_M(smt)(f) \quad (s, t \in S, m \in M, f \in M^*),$$

which proves bimodule compatibility.  $\square$

### 1.13 Exercises

**Exercise 1.50 (Categories and shifts).** Let  $A = \mathbb{k}A_2$  with the grading of Example 1.13. Show that  $e_{11}A$  and  $(e_{11}A)(1)$  are isomorphic in  $\text{Mod } A$  but not in  $\text{GrMod } A$ . Hint. Compare their degree 1 components. Explain why this does not contradict the fact that the shift functor is an equivalence.

**Exercise 1.51 (Matrix calculations).** Verify the grading on  $\mathbb{k}A_n$  in Example 1.13 using  $e_{ij}e_{pq} = \delta_{jp}e_{iq}$ . Compute  $\text{rad}(\mathbb{k}A_3)$  from the definition. Hint. The strictly upper triangular ideal  $J$  satisfies  $J^3 = 0$ . Apply the argument of Example 1.11 to simple modules and to  $A/J \cong \mathbb{k}^{\oplus 3}$ .

**Exercise 1.52 (Shifts and duals).** For  $A = \mathbb{k}A_3$ , compute the degrees and all non-zero matrix-unit actions on  $\mathbb{D}(e_{22}A)$ . Show that it is not isomorphic to any left module  $Ae_{jj}$ , even after a shift. Hint. Compare dimensions and the actions of  $e_{11}, e_{22}, e_{33}$ .

**Exercise 1.53 (Homogeneous maps).** For the map  $f$  in Example 1.23, write every non-zero homogeneous component  $f_d$  explicitly and prove that no finite sum equals  $f$ .

**Exercise 1.54 (Dependence on the form).** Repeat Example 1.48 with  $\tau(e_i) = c_i \neq 0$ . For  $V = \mathbb{k}e_{12}$ , find the right and left  $S$ -linear maps corresponding to the functional  $\varepsilon(e_{12}) = 1$  under (1.9). Check. Their values on  $e_{12}$  are  $c_2^{-1}e_2$  and  $c_1^{-1}e_1$ , respectively.

## 2 Tensor and quadratic algebras

### 2.1 Tensor products over an algebra

Let  $A$  be a graded algebra and set  $S = A_0$ . Multiplication on  $A_1$  satisfies  $(vs)w = v(sw)$  for  $v, w \in A_1$  and  $s \in S$ . A tensor product over  $S$  incorporates this identity.

**Definition 2.1.** For a  $\mathbb{k}$ -algebra  $S$ , a right  $S$ -module  $M$ , and a left  $S$ -module  $N$ , the **tensor product**  $M \otimes_S N$  is the quotient of  $M \otimes_{\mathbb{k}} N$  by

$$\text{span}_{\mathbb{k}}\{ms \otimes n - m \otimes sn \mid m \in M, n \in N, s \in S\}.$$

Write  $m \otimes_S n$  for the image of  $m \otimes n$ . No semisimplicity or finiteness assumption on  $S, M, N$  is needed for the following properties.

**Lemma 2.2 (Basic properties of tensor products).** Let  $S$  be a  $\mathbb{k}$ -algebra,  $M \in \text{Mod } S$ , and  $N \in \text{Mod}(S^{\text{op}})$ .

- (1) For a vector space  $W$  and a bilinear map  $b : M \times N \rightarrow W$  satisfying  $b(ms, n) = b(m, sn)$  for all  $m \in M, s \in S$ , and  $n \in N$ , there is a unique linear map  $\tilde{b}$  making the diagram commute:

$$\begin{array}{ccc} M \times N & \xrightarrow{(m,n) \mapsto m \otimes_S n} & M \otimes_S N \\ & \searrow b & \downarrow \exists! \tilde{b} \\ & & W. \end{array}$$

Conversely, every linear map  $M \otimes_S N \rightarrow W$  defines such a bilinear map.

- (2) If  $M, N$  are  $S$ -bimodules, then  $M \otimes_S N$  is an  $S$ -bimodule with  $s(m \otimes_S n)t = sm \otimes_S nt$  for  $s, t \in S$ .  
(3) For  $S$ -bimodules  $U, V, W$ , there are bimodule isomorphisms

$$\begin{aligned} (U \otimes_S V) \otimes_S W &\xrightarrow{\sim} U \otimes_S (V \otimes_S W), & (u \otimes v) \otimes w &\mapsto u \otimes (v \otimes w), \\ S \otimes_S V &\xrightarrow{\sim} V, & s \otimes v &\mapsto sv, \\ V \otimes_S S &\xrightarrow{\sim} V, & v \otimes s &\mapsto vs. \end{aligned}$$

- (4) Tensor products distribute over direct sums in either variable. For a family  $(M_i)_{i \in I}$  of right  $S$ -modules, the isomorphism is

$$\left( \bigoplus_{i \in I} M_i \right) \otimes_S N \xrightarrow{\sim} \bigoplus_{i \in I} (M_i \otimes_S N), \quad (m_i)_i \otimes n \mapsto (m_i \otimes n)_i.$$

**Proof** For (1), Definition 1.30 gives a unique linear map  $M \otimes_{\mathbb{k}} N \rightarrow W$  sending  $m \otimes n$  to  $b(m, n)$ . It descends to the quotient precisely when  $b(ms, n) - b(m, sn) = 0$ . Uniqueness follows because the elements  $m \otimes_S n$  span the quotient.

For (2), the outer actions preserve the defining relations: for  $a, s, t \in S$ ,

$$s(ma \otimes n - m \otimes an)t = (sm)a \otimes nt - sm \otimes a(nt).$$

The module axioms follow from those on  $M, N$ . In (3), both parenthesisations impose the relations that move an element of  $S$  between adjacent factors, so the displayed associativity map and its reverse are well-defined. Their composites fix every  $(u \otimes v) \otimes w$  and  $u \otimes (v \otimes w)$ . The unit inverses are  $V \rightarrow S \otimes_S V, v \mapsto 1 \otimes v$ , and  $V \rightarrow V \otimes_S S, v \mapsto v \otimes 1$ ; the tensor relations give the required identities.

For (4), the displayed map has finite support because  $(m_i)_i$  does. It is well-defined by (1). If  $\iota_i : M_i \rightarrow \bigoplus_{j \in I} M_j$  is the canonical inclusion, the inverse sends  $m \otimes n$  in the  $i$ th summand to  $\iota_i(m) \otimes n$  and extends linearly over the direct sum. Both composites fix the displayed generators. The same construction works in the other variable.  $\square$

For right module maps  $u : M \rightarrow M'$  and left module maps  $v : N \rightarrow N'$ , the universal property defines

$$u \otimes_S v : M \otimes_S N \longrightarrow M' \otimes_S N', \quad m \otimes n \longmapsto u(m) \otimes v(n).$$

These maps satisfy

$$(u' \otimes v') \circ (u \otimes v) = (u' \circ u) \otimes (v' \circ v), \quad \text{id}_M \otimes \text{id}_N = \text{id}_{M \otimes_S N}$$

whenever the maps are composable. Thus, for fixed  $N$  and fixed  $M$ , respectively, tensor product gives functors

$$(-) \otimes_S N : \text{Mod } S \longrightarrow \text{Mod } \mathbb{k}, \quad M \otimes_S (-) : \text{Mod}(S^{\text{op}}) \longrightarrow \text{Mod } \mathbb{k}.$$

For bimodules and bimodule maps, the same construction preserves the outer actions in Lemma 2.2(2).

The equality  $ms \otimes_S n = m \otimes_S sn$  allows a scalar from  $S$  to pass between adjacent factors. For an  $S$ -bimodule  $V$ , define

$$V^{\otimes_S 0} := S, \quad V^{\otimes_S i} := \underbrace{V \otimes_S \cdots \otimes_S V}_{i \text{ factors}} \quad (i \geq 1).$$

Lemma 2.2 makes these  $S$ -bimodules independent of parenthesisation and supplies the identifications used in concatenation.

For an  $S$ -bimodule  $M$ , the right and left  $S$ -duals of Definition 1.39 are themselves  $S$ -bimodules, with actions

$$(sft)(m) = sf(tm), \quad (sgt)(m) = g(ms)t \quad (s, t \in S, m \in M). \quad (2.1)$$

Here  $f \in M^*$  and  $g \in {}^*M$ . The duals are ungraded; the parenthesised stars of Definition 1.45 denote different, graded constructions.

Tensor duality reverses the order of the factors.

**Proposition 2.3.** *Let  $S$  be finite-dimensional semisimple and let  $V, W$  be finite-dimensional  $S$ -bimodules. There are bimodule isomorphisms, natural in  $V, W$ ,*

$$\begin{aligned} \theta_r : W^* \otimes_S V^* &\xrightarrow{\sim} (V \otimes_S W)^*, \\ \theta_\ell : {}^*W \otimes_S {}^*V &\xrightarrow{\sim} {}^*(V \otimes_S W), \end{aligned} \quad (2.2)$$

defined by

$$\theta_r(g \otimes f)(v \otimes w) = g(f(v)w), \quad \theta_\ell(g \otimes f)(v \otimes w) = f(vg(w)). \quad (2.3)$$

**Proof** *Well-definedness.* For the right duals, right  $S$ -linearity gives

$$g(f(vs)w) = g(f(v)sw), \quad (gs)(f(v)w) = g(sf(v)w) = g((sf)(v)w).$$

The first identity respects the relation in  $V \otimes_S W$ , and the second respects the relation in  $W^* \otimes_S V^*$ . The resulting functional is right  $S$ -linear because  $g$  is. For the left duals, the corresponding identities are

$$f(vs g(w)) = f(vg(sw)), \quad f(v(gs)(w)) = f(vg(w)s) = (sf)(vg(w)).$$

Left  $S$ -linearity of  $f$  makes the resulting functional left  $S$ -linear. The outer actions in (2.1) give

$$\begin{aligned} \theta_r(sg \otimes ft)(v \otimes w) &= sg(f(tv)w), \\ \theta_\ell(sg \otimes ft)(v \otimes w) &= f(vg(ws))t, \end{aligned}$$

which are the required bimodule identities.

*Bijectivity.* To check  $\theta_r$ , forget the left action on  $V$ . For  $V = S$ , identify  $S^* \cong S$  by  $f \mapsto f(1)$ ; then  $\theta_r$  is  $W^* \otimes_S S \rightarrow W^*$ ,  $g \otimes s \mapsto gs$ , an isomorphism by Lemma 2.2(3). The formula commutes with maps of right  $S$ -modules in  $V$ , so it is an isomorphism for finite direct sums and direct summands of  $S$ . Corollary 1.10 applies to  $V$ . For  $\theta_\ell$ , use  $W = S$  as a left module, where the map is  $S \otimes_S {}^*V \rightarrow {}^*V$ ,  $s \otimes f \mapsto sf$ , and apply the same argument to  $W$ .

*Naturality.* For bimodule maps  $p : V \rightarrow V'$  and  $q : W \rightarrow W'$ , the square

$$\begin{array}{ccc} (W')^* \otimes_S (V')^* & \xrightarrow{\theta_r} & (V' \otimes_S W')^* \\ q^* \otimes p^* \downarrow & & \downarrow (p \otimes q)^* \\ W^* \otimes_S V^* & \xrightarrow{\theta_r} & (V \otimes_S W)^* \end{array}$$

commutes: both routes send  $g \otimes f$ , evaluated at  $v \otimes w$ , to  $g(f(p(v))q(w))$ . For  $\theta_\ell$ , the corresponding two routes give  $f(p(v)g(q(w)))$ .  $\square$

Neither map in (2.2) requires a symmetrising form. These canonical identifications are the reason for using  $S$ -duals in tensor calculations.

For the ungraded  $\mathbb{k}$ -linear dual  $D$ , there is also a natural bimodule isomorphism [GI20, §2.1]

$$\begin{aligned} \theta_D : D(W) \otimes_S V^* &\xrightarrow{\sim} D(V \otimes_S W), \\ \theta_D(\lambda \otimes f)(v \otimes w) &= \lambda(f(v)w). \end{aligned}$$

The actions on  $D(W)$  are those in (1.8). The tensor relations are checked as for  $\theta_r$ . For  $V = S$ , the map is  $D(W) \otimes_S S \rightarrow D(W)$ ,  $\lambda \otimes s \mapsto \lambda s$ ; the direct summand argument in Proposition 2.3 proves bijectivity.

Suppose now that  $S$  is equipped with a symmetrising form  $\tau : S \rightarrow \mathbb{k}$ , as in Definition 1.41. The two-sided dual of Definition 1.40 satisfies

$$\begin{aligned} \theta_{\text{en}} : W^* \otimes_S V^{*\text{en}} &\xrightarrow{\sim} (V \otimes_S W)^{*\text{en}}, \\ \theta_{\text{en}}(g \otimes F)(v \otimes w) &= \sum_j g(a_j w) \otimes b_j^{\text{op}}, \quad F(v) = \sum_j a_j \otimes b_j^{\text{op}}. \end{aligned} \quad (2.4)$$

The formula respects the tensor relations and both outer actions. For example,  $F(vs) = F(v)(s \otimes 1)$  replaces  $a_j$  by  $a_j s$ , while  $(Fs)(v) = (1 \otimes s^{\text{op}})F(v)$  replaces  $b_j$  by  $b_j s$ . These give, respectively, compatibility with  $vs \otimes w = v \otimes sw$  and with the right outer action. Compatibility with  $(gs) \otimes F = g \otimes (sF)$  follows from  $(gs)(a_j w) = g(sa_j w)$ . The other actions follow from right  $S$ -linearity of  $g$  and  $F(tv) = F(v)(1 \otimes t^{\text{op}})$ .

The comparison maps  $\gamma_M$  in (1.11) give

$$\gamma_{V \otimes_S W} \circ \theta_{\text{en}} = \theta_r \circ (\text{id}_{W^*} \otimes \gamma_V),$$

since both sides send  $g \otimes F$ , evaluated at  $v \otimes w$ , to  $\sum_j g(a_j w) \tau(b_j)$ . Thus  $\theta_{\text{en}}$  is an isomorphism by Proposition 1.43. Precomposition shows that  $\theta_D$  and  $\theta_{\text{en}}$  are natural. No assumption that  $S^{\text{en}}$  is semisimple is used; compare Remark 1.32.

For the maps  $\alpha_M(f) = \tau \circ f$  in (1.9),

$$\alpha_{V \otimes_S W} \circ \theta_r = \theta_D \circ (\alpha_W \otimes \text{id}_{V^*}),$$

because both sides send  $g \otimes f$  to the functional  $v \otimes w \mapsto \tau(g(f(v)w))$ .

**Example 2.4.** Let  $S = \mathbb{k}^{\oplus r}$  with coordinate idempotents  $e_1, \dots, e_r$ , and suppose  $V = e_i V e_j$  and  $W = e_j W e_\ell$ . By (2.1),

$$V \otimes_S W = e_i(V \otimes_S W)e_\ell, \quad W^* \otimes_S V^* = e_\ell(W^* \otimes_S V^*)e_i.$$

Thus  $\theta_r$  in (2.2) reverses the two outer idempotents.

## 2.2 Generated algebras

Let  $A$  be graded with  $S = A_0$ . Multiplication makes  $A_1$  an  $S$ -bimodule and gives

$$\mu_i : A_1^{\otimes_S i} \longrightarrow A_i, \quad a_1 \otimes \dots \otimes a_i \longmapsto a_1 \dots a_i \quad (i \geq 1), \quad \mu_0 = \text{id}_S. \quad (2.5)$$

**Definition 2.5.** The algebra  $A$  is *generated in degree 1 over  $S$*  if every  $\mu_i$  in (2.5) is surjective.

Equivalently,  $S$  and  $A_1$  generate  $A$  as an algebra.

**Example 2.6.** The algebra  $\mathbb{k}A_n$  of Example 1.13 is generated in degree 1 over  $S = \bigoplus_i \mathbb{k}e_{ii}$ :

$$e_{ij} = e_{i,i+1}e_{i+1,i+2} \dots e_{j-1,j} \quad (1 \leq i < j \leq n).$$

For the grading  $\deg e_{ij} = 2(j - i)$ , the same algebra has  $A_1 = 0$  and  $A_2 \neq 0$ , so it is not generated in degree 1.

## 2.3 Tensor algebras and the universal property

**Definition 2.7.** Let  $S$  be a  $\mathbb{k}$ -algebra and  $V$  an  $S$ -bimodule. The *tensor algebra of  $V$  over  $S$*  is

$$T_S(V) := \bigoplus_{i \in \mathbb{Z}} T_S(V)_i, \quad \text{where} \quad T_S(V)_i = \begin{cases} V^{\otimes_S i}, & i \geq 1, \\ S, & i = 0, \\ 0, & i < 0, \end{cases}$$

with multiplication given by concatenation of tensors and the given  $S$ -actions.

Every element of  $T_S(V)$  is a finite sum of tensors, since the grading is a direct sum. The unit is  $1_S \in T_S(V)_0$ . Associativity follows from the associativity and unit isomorphisms in Lemma 2.2(3). The left and right  $S$ -actions determine which factors can be concatenated. Juxtaposition  $v_1 \dots v_d$  in  $T_S(V)$  denotes  $v_1 \otimes_S \dots \otimes_S v_d$ ; in a specified quotient it denotes the image of that tensor.

**Lemma 2.8.** Let  $S = \bigoplus_{i=1}^r \mathbb{k}e_i$  with coordinate idempotents and let  $V, W$  be  $S$ -bimodules. There is a bimodule isomorphism

$$V \otimes_S W \cong \bigoplus_{i,j,\ell} e_i V e_j \otimes_{\mathbb{k}} e_j W e_\ell. \quad (2.6)$$

Consequently, after choosing a basis of each  $e_i V e_j$ , a basis of  $V^{\otimes_S d}$  for  $d \geq 1$  consists of the words of length  $d$  whose adjacent idempotents match.

**Proof** The mutually orthogonal idempotents sum to 1, so

$$V = \bigoplus_{i,j} e_i V e_j, \quad (e_i V e_j) \otimes_S (e_p W e_\ell) = 0 \quad \text{if } j \neq p. \quad (2.7)$$

For the second identity,  $v \otimes w = ve_j \otimes w = v \otimes e_j w = 0$ . The inclusions of the matching components define the map from right to left in (2.6). Its inverse is

$$V \otimes_S W \longrightarrow \bigoplus_{i,j,\ell} e_i V e_j \otimes_{\mathbb{k}} e_j W e_\ell, \quad v \otimes_S w \longmapsto \sum_{i,j,\ell} e_i v e_j \otimes_{\mathbb{k}} e_j w e_\ell.$$

For  $s = \sum_j s_j e_j \in S$ , replacing  $(v, w)$  by  $(vs, w)$  or  $(v, sw)$  multiplies the summands with middle index  $j$  by the same scalar  $s_j$ , so the inverse is well-defined. One composite sends  $v \otimes_S w$  to  $\sum_{i,j,\ell} e_i v e_j \otimes_S e_j w e_\ell = v \otimes_S w$ . On  $v \otimes_{\mathbb{k}} w$  with  $v \in e_i V e_j$  and  $w \in e_j W e_\ell$ , the other composite has only the  $(i, j, \ell)$  component, equal to  $v \otimes_{\mathbb{k}} w$ . Iterating (2.6) and using the basis construction following Definition 1.30 proves the assertion about words.  $\square$

**Example 2.9.** Let  $S = \mathbb{k}^{\oplus 3}$  with coordinate idempotents  $e_1, e_2, e_3$ , and let  $V = \mathbb{k}a \oplus \mathbb{k}b$  have actions  $e_1 a e_2 = a$  and  $e_2 b e_3 = b$ , with all other idempotent actions zero. By Lemma 2.8,

$$V \otimes_S V = \mathbb{k}(a \otimes b), \quad V^{\otimes_S 3} = 0, \quad T_S(V) = S \oplus \mathbb{k}a \oplus \mathbb{k}b \oplus \mathbb{k}ab.$$

Here  $ab = a \otimes b$ . The assignments  $e_i \mapsto e_{ii}$ ,  $a \mapsto e_{12}$ , and  $b \mapsto e_{23}$  give  $T_S(V) \cong \mathbb{k}A_3$ ; the tensor  $ab$  maps to  $e_{13}$ .

**Proposition 2.10 (Universal property of the tensor algebra).** Let  $\iota_S : S \rightarrow T_S(V)$  and  $\iota_V : V \rightarrow T_S(V)$  be the canonical inclusions. Given a unital  $\mathbb{k}$ -algebra homomorphism  $\varphi : S \rightarrow B$  and an  $S$ -bimodule map  $f : V \rightarrow B$ , where  $sbt = \varphi(s)b\varphi(t)$  on  $B$ , there is a unique  $\mathbb{k}$ -algebra homomorphism  $\tilde{f} : T_S(V) \rightarrow B$  making the diagram commute:

$$\begin{array}{ccccc} V & \xrightarrow{\iota_V} & T_S(V) & \xleftarrow{\iota_S} & S \\ & \searrow f & \downarrow \exists! \tilde{f} & \swarrow \varphi & \\ & & B & & \end{array} \quad (2.8)$$

**Proof** Define  $\tilde{f}$  on  $S$  and each tensor power by

$$\tilde{f}(s) = \varphi(s), \quad \tilde{f}(v_1 \otimes_S \cdots \otimes_S v_i) = f(v_1) \cdots f(v_i) \quad (s \in S, i \geq 1). \quad (2.9)$$

The identity  $f(vs)f(w) = f(v)\varphi(s)f(w) = f(v)f(sw)$  respects the tensor relations. Since every element of  $T_S(V)$  has finitely many non-zero components, these formulas extend to a linear map  $T_S(V) \rightarrow B$ . Concatenation of tensors becomes multiplication in  $B$ , and the identities  $f(sv) = \varphi(s)f(v)$  and  $f(vs) = f(v)\varphi(s)$  handle the degree 0 factors. Also,  $\tilde{f}(1_S) = \varphi(1_S) = 1_B$ . Every algebra homomorphism making (2.8) commute must have the values in (2.9), proving uniqueness.  $\square$

If  $B$  is graded,  $\varphi(S) \subset B_0$ , and  $f(V) \subset B_1$ , then (2.9) gives  $\tilde{f}(V^{\otimes_S i}) \subset B_i$ . Thus  $\tilde{f}$  is a graded algebra homomorphism.

**Corollary 2.11.** If  $A$  is generated in degree 1 over  $S = A_0$ , there is a surjective graded algebra homomorphism  $\mu : T_S(A_1) \twoheadrightarrow A$  whose degree  $i$  component is  $\mu_i$  from (2.5).

**Proof** Apply Proposition 2.10 to the inclusions  $S \rightarrow A$  and  $A_1 \rightarrow A$ . Surjectivity of each  $\mu_i$  gives surjectivity of  $\mu$ .  $\square$

**Definition 2.12.** For the map  $\mu$  in Corollary 2.11, the homogeneous two-sided ideal  $I := \text{Ker } \mu$  is the *ideal of relations*.

Writing  $V = A_1$ , the components  $\mu_0 = \text{id}_S$  and  $\mu_1 = \text{id}_V$  give  $I_0 = I_1 = 0$ . Since  $\mu$  preserves degrees, its kernel is homogeneous, and

$$T_S(V)/I \xrightarrow{\sim} A, \quad z + I \longmapsto \mu(z)$$

is a graded algebra isomorphism. The bimodule  $V$  specifies the generators and their left and right  $S$ -actions;  $I$  specifies the relations between their products. Homogeneity is essential for the quotient grading: the relation  $x - x^2$  in Example 1.28 does not satisfy this condition.

## 2.4 Quadratic algebras

For quadratic presentations, let  $S$  be finite-dimensional semisimple,  $V$  a finite-dimensional  $S$ -bimodule, and  $R \subset V \otimes_S V$  an  $S$ -sub-bimodule. Write  $(R)$  for the two-sided ideal generated by  $R$  in  $T_S(V)$ , with  $V$  in degree 1.

**Definition 2.13.** A graded algebra  $A$  is *quadratic* if  $A \cong T_S(V)/(R)$  as graded algebras for some  $S, V, R$  satisfying the preceding conditions.

**Proposition 2.14.** A quadratic algebra  $A = T_S(V)/(R)$  has  $A_0 = S$ ,  $A_1 = V$ , and  $A_2 = (V \otimes_S V)/R$ , is locally finite, and is generated in degree 1 over  $S$ .

**Proof** The ideal  $(R)$  has no components in degrees 0 and 1. Its degree 2 component is exactly  $R$ . Each  $A_i$  is a quotient of the finite-dimensional space  $V^{\otimes_S i}$ , and multiplication maps  $V^{\otimes_S i}$  onto  $A_i$ .  $\square$

Thus a quadratic presentation recovers  $S \cong A_0$ ,  $V \cong A_1$ , and  $R \cong \text{Ker}(A_1 \otimes_{A_0} A_1 \rightarrow A_2)$ . In particular, the kernel of the canonical map  $T_{A_0}(A_1) \rightarrow A$  is generated by its degree 2 component. Quadraticity depends on the grading: for  $n \geq 2$ , the grading  $\deg e_{ij} = 2(j-i)$  on  $\mathbb{k}A_n$  is not quadratic, since it has no degree 1 generators.

**Example 2.15.** Let  $S = \bigoplus_{i=1}^n \mathbb{k}e_{ii}$  and  $V = \bigoplus_{i=1}^{n-1} \mathbb{k}e_{i,i+1}$  inside  $\mathbb{k}A_n$ . By Lemma 2.8, the basis tensors of positive length are

$$e_{i,i+1} \otimes_S e_{i+1,i+2} \otimes_S \cdots \otimes_S e_{j-1,j} \mapsto e_{ij} \quad (1 \leq i < j \leq n).$$

Thus multiplication gives  $T_S(V) \cong \mathbb{k}A_n$ , a quadratic algebra with  $R = 0$ .

**Example 2.16.** For  $S, V, a, b$  from Example 2.9, the quotient

$$T_S(V)/(ab) \cong \mathbb{k}A_3/(e_{13})$$

is quadratic with  $R = \mathbb{k}(a \otimes b)$ . Its basis is  $e_1, e_2, e_3, a, b$ , in degrees 0, 0, 0, 1, 1, as in Example 1.27.

**Example 2.17.** Let  $S = \mathbb{k}^{\oplus 2}$  with coordinate idempotents  $e_1, e_2$ , and let  $V = \mathbb{k}a \oplus \mathbb{k}b$  satisfy  $e_1 a e_2 = a$  and  $e_2 b e_1 = b$ , with all other idempotent actions zero. By Lemma 2.8, the alternating words in  $a, b$  form the basis of positive degree tensors in  $T_S(V)$ . The quadratic algebra  $A = T_S(V)/(ab)$  has

$$A = S \oplus \mathbb{k}a \oplus \mathbb{k}b \oplus \mathbb{k}ba, \quad \dim_{\mathbb{k}} A_0 = 2, \quad \dim_{\mathbb{k}} A_1 = 2, \quad \dim_{\mathbb{k}} A_2 = 1.$$

Every alternating word of length at least 3 contains  $ab$ , so all higher components vanish.

**Example 2.18.** Let  $B = \mathbb{k}A_4$  and  $A = B/(e_{14})$ , with the grading of Example 1.13. The ideal  $I = (e_{14}) = \mathbb{k}e_{14}$  has  $I_2 = 0$  but  $I_3 \neq 0$ . Since  $e_{14} = e_{12}e_{23}e_{34}$ , the algebra  $A$  is generated in degree 1 with a relation of degree 3 and no relations of degree 2. By Proposition 2.14, any quadratic presentation would have the same generators and zero quadratic relation space, hence would give  $\mathbb{k}A_4$  itself. Thus  $A$  is not quadratic.

**Example 2.19 (Polynomial and exterior algebras).** For the classical comparison, take  $S = \mathbb{k}$  and  $V = \bigoplus_{i=1}^n \mathbb{k}x_i$ . The *symmetric algebra* and *exterior algebra* are

$$\begin{aligned} \text{Sym}(V) &= T_{\mathbb{k}}(V)/(x_i x_j - x_j x_i \mid i < j) \cong \mathbb{k}[x_1, \dots, x_n], \\ \bigwedge V &= T_{\mathbb{k}}(V)/(v^2 \mid v \in V). \end{aligned} \tag{2.10}$$

Both are quadratic. The polynomial basis allows repetitions; the exterior basis consists of 1 and  $x_{i_1} \cdots x_{i_d}$  with  $i_1 < \cdots < i_d$ . The exterior relations are equivalently  $x_i^2 = 0$  for  $1 \leq i \leq n$  and  $x_i x_j + x_j x_i = 0$  for  $i < j$ , in every characteristic.

The bases in Example 2.19 can be checked directly. For  $\text{Sym}(V)$ , commuting adjacent generators gives the ordered monomials, which are independent under the map to the polynomial ring. For  $\bigwedge V$ , the relations reduce every word to a multiple of a strictly increasing word or to zero. To prove independence in degree  $d$ , fix  $i_1 < \dots < i_d$  and send  $v_1 \otimes \dots \otimes v_d$  to the determinant of their coordinates in rows  $i_1, \dots, i_d$ . This functional vanishes on tensors with two equal adjacent factors, hence on the degree  $d$  part of  $(v^2 \mid v \in V)$ . On increasing basis words it is 1 on  $x_{i_1} \dots x_{i_d}$  and 0 on the others, proving independence in every characteristic.

## 2.5 Hilbert series

A Hilbert series records the dimension of each homogeneous component.

**Definition 2.20.** Let  $M$  be a locally finite graded vector space such that  $M_j = 0$  for all  $j < j_0$ , for some  $j_0 \in \mathbb{Z}$ . Its *Hilbert series* is the formal Laurent series  $h_M(t) := \sum_{j \in \mathbb{Z}} (\dim_{\mathbb{k}} M_j) t^j$ .

For a locally finite graded algebra  $A$ ,  $h_A(t)$  refers to its underlying graded vector space; the series does not record multiplication. Its constant coefficient is  $\dim_{\mathbb{k}} A_0$ , which need not be 1.

**Lemma 2.21.** Let  $L, M, N$  be locally finite graded vector spaces concentrated in degrees bounded below.

- (1) A graded isomorphism  $M \cong N$  implies  $h_M(t) = h_N(t)$ .
- (2) A short exact sequence  $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$  in  $\text{grmod } \mathbb{k}$  gives  $h_M(t) = h_L(t) + h_N(t)$ .
- (3) For every  $d \in \mathbb{Z}$ ,  $h_{M(d)}(t) = t^{-d} h_M(t)$ .

**Proof** A graded isomorphism restricts to an isomorphism in each degree. By Definition 1.36, the sequence in (2) is exact in each degree, so  $\dim_{\mathbb{k}} M_j = \dim_{\mathbb{k}} L_j + \dim_{\mathbb{k}} N_j$  for every  $j \in \mathbb{Z}$ . For shifts, Definition 1.20 gives

$$h_{M(d)}(t) = \sum_j \dim_{\mathbb{k}}(M_{j+d}) t^j = t^{-d} \sum_i \dim_{\mathbb{k}}(M_i) t^i. \quad \square$$

**Example 2.22.** For the upper triangular algebra in Example 1.13, counting matrix entries on each diagonal gives

$$h_{\mathbb{k}A_n}(t) = n + (n-1)t + \dots + t^{n-1}, \quad h_{\mathbb{k}A_3/(e_{13})}(t) = 3 + 2t.$$

The second equality also follows by subtracting  $h_{(e_{13})}(t) = t^2$  from  $h_{\mathbb{k}A_3}(t)$ , as in Lemma 2.21(2). For  $A = T_S(V)/(ab)$  in Example 2.17, the basis  $e_1, e_2, a, b, ba$  gives  $h_A(t) = 2 + 2t + t^2$ .

**Example 2.23.** For the algebras in Example 2.19, each polynomial generator may occur any number of times, whereas each exterior generator occurs at most once. Thus

$$h_{\text{Sym}(V)}(t) = \frac{1}{(1-t)^n}, \quad h_{\bigwedge V}(t) = (1+t)^n \quad \text{where } n = \dim_{\mathbb{k}} V.$$

**Definition 2.24.** For graded vector spaces  $V, W$ , the *total grading* on their tensor product is

$$(V \otimes_{\mathbb{k}} W)_n = \bigoplus_{i+j=n} V_i \otimes_{\mathbb{k}} W_j.$$

The decomposition in Definition 2.24 follows from Lemma 2.2(4), applied over  $\mathbb{k}$ :

$$\left( \bigoplus_i V_i \right) \otimes_{\mathbb{k}} \left( \bigoplus_j W_j \right) \cong \bigoplus_{i,j} (V_i \otimes_{\mathbb{k}} W_j).$$

An individual tensor has finitely many non-zero components, even if the sum defining a fixed total degree has infinitely many summands.

**Lemma 2.25.** *For locally finite graded vector spaces  $V, W$  concentrated in degrees bounded below,  $V \otimes_{\mathbb{k}} W$  is locally finite and bounded below, and  $h_{V \otimes_{\mathbb{k}} W}(t) = h_V(t)h_W(t)$ .*

**Proof** Choose integers  $a, b$  with  $V_i = 0$  for  $i < a$  and  $W_j = 0$  for  $j < b$ . For total degree  $n$ , only  $a \leq i \leq n - b$  can contribute, so

$$\dim_{\mathbb{k}}(V \otimes_{\mathbb{k}} W)_n = \sum_{i+j=n} \dim_{\mathbb{k}} V_i \dim_{\mathbb{k}} W_j$$

is a finite sum and is zero for  $n < a + b$ . The sum is precisely the coefficient of  $t^n$  in  $h_V(t)h_W(t)$ .  $\square$

## 2.6 The degree 2 relations of a quotient

Let  $S$  be finite-dimensional semisimple,  $V$  a finite-dimensional  $S$ -bimodule, and  $I \subset T_S(V)$  a homogeneous two-sided ideal with  $I_0 = I_1 = 0$ . Set  $A = T_S(V)/I$ . In degree 2,

$$A_2 \cong (V \otimes_S V)/I_2, \quad \text{where } I_2 = I \cap (V \otimes_S V).$$

**Definition 2.26.** *The **quadratic part** of  $A$  is  $qA := T_S(V)/(I_2)$ .*

Since  $(I_2) \subset I$ , there is a surjective graded algebra homomorphism

$$\pi_A : qA \longrightarrow A, \quad z + (I_2) \longmapsto z + I, \quad \text{Ker } \pi_A = I/(I_2).$$

Consequently,

$$\pi_A \text{ is an isomorphism} \iff I = (I_2) \iff A \text{ is quadratic.}$$

For the last equivalence,  $(I_2)$  gives a quadratic presentation, while Proposition 2.14 identifies the generators and degree 2 relations of any quadratic presentation with  $A_1$  and  $I_2$ .

**Example 2.27.** *For  $A = \mathbb{k}A_4/(e_{14})$  from Example 2.18, we have  $I_2 = 0$ , so*

$$qA = \mathbb{k}A_4, \quad \pi_A : \mathbb{k}A_4 \rightarrow \mathbb{k}A_4/(e_{14}), \quad \text{Ker } \pi_A = \mathbb{k}e_{14}.$$

*Thus  $qA$  does not impose the cubic relation  $e_{12}e_{23}e_{34} = 0$ .*

## 2.7 Orthogonal complements and quadratic duality

Let  $S$  be finite-dimensional semisimple.

**Definition 2.28.** *For an  $S$ -sub-bimodule  $U$  of a finite-dimensional  $S$ -bimodule  $M$ , its **right orthogonal complement** and **left orthogonal complement** are*

$$\begin{aligned} U^\perp &:= \{f \in M^* \mid f(u) = 0 \text{ for all } u \in U\}, \\ {}^\perp U &:= \{g \in {}^*M \mid g(u) = 0 \text{ for all } u \in U\}. \end{aligned}$$

Both orthogonal complements are  $S$ -sub-bimodules. Indeed, for  $s, t \in S$  and  $u \in U$ , (2.1) gives

$$(sft)(u) = sf(tu) = 0 \quad (f \in U^\perp), \quad (sgt)(u) = g(us)t = 0 \quad (g \in {}^\perp U),$$

since  $tu, us \in U$ . The orthogonal complements are defined without choosing a symmetrising form.

If  $S$  is equipped with a symmetrising form  $\tau : S \rightarrow \mathbb{k}$ , the comparison maps  $\alpha_M, \beta_M$  of (1.9) satisfy

$$\alpha_M(U^\perp) = \beta_M({}^\perp U) = \{\lambda \in D(M) \mid \lambda(u) = 0 \text{ for all } u \in U\}. \quad (2.11)$$

The forward inclusions follow by composing with  $\tau$ . Conversely,  $\alpha_M(f)(U) = 0$  gives

$$\tau(f(u)s) = \tau(f(us)) = 0 \quad \text{for all } s \in S, u \in U.$$

Non-degeneracy of  $\tau$  forces  $f(u) = 0$  for every  $u \in U$ . For the left dual,  $\tau(sg(u)) = \tau(g(su)) = 0$  gives  $g(u) = 0$  instead. Bijectivity of  $\alpha_M, \beta_M$  in Proposition 1.43 proves (2.11).

**Example 2.29.** For  $\text{Sym}(V)$  and  $\bigwedge V$  in Example 2.19, take  $V = \mathbb{k}x \oplus \mathbb{k}y$ . The relation spaces are

$$\begin{aligned} R_{\text{Sym}(V)} &= \mathbb{k}(x \otimes y - y \otimes x), \\ R_{\bigwedge V} &= \mathbb{k}(x \otimes x) \oplus \mathbb{k}(y \otimes y) \oplus \mathbb{k}(x \otimes y + y \otimes x). \end{aligned}$$

Let  $x^*, y^* \in V^*$  be the coordinate functionals. Identifying  $(V \otimes_{\mathbb{k}} V)^*$  with  $V^* \otimes_{\mathbb{k}} V^*$  using  $\theta_r$  from (2.3),

$$\begin{aligned} R_{\text{Sym}(V)}^\perp &= \text{span}_{\mathbb{k}}\{x^* \otimes x^*, y^* \otimes y^*, x^* \otimes y^* + y^* \otimes x^*\}, \\ R_{\bigwedge V}^\perp &= \mathbb{k}(x^* \otimes y^* - y^* \otimes x^*). \end{aligned}$$

Thus taking orthogonal complements exchanges the symmetric and exterior relation spaces.

## 2.8 Exercises

**Exercise 2.30.** Let  $A = \mathbb{k}A_4/(e_{13}, e_{24})$  with the grading of Example 1.13. Find a homogeneous basis, compute  $h_A(t)$ , and write  $A = T_S(V)/(R)$  with explicit  $S, V, R$ . Hint. Use Lemma 2.8; the ideal contains  $e_{14}$  as well as  $e_{13}, e_{24}$ . The Hilbert series is  $4 + 3t$ .

**Exercise 2.31.** For the bimodule  $V$  in Example 2.17, find a basis of each  $T_S(V)_d$  and show that  $h_{T_S(V)}(t) = 2/(1-t)$ . Compare with the quotient  $T_S(V)/(ab)$ . Hint. Count alternating words separately in each degree, including degree 0.

**Exercise 2.32.** For the algebras in Example 2.19, prove

$$\dim_{\mathbb{k}} \text{Sym}(V)_2 = \frac{n(n+1)}{2}, \quad \dim_{\mathbb{k}} (\bigwedge V)_2 = \frac{n(n-1)}{2} \quad \text{where } n = \dim_{\mathbb{k}} V.$$

Hint. Count the ordered pairs allowed in the two bases of Example 2.19.

### 3 Lecture 3: The quadratic dual

#### 3.1 Generators and orthogonal relations

Throughout this lecture,  $S$  is a finite-dimensional semisimple  $\mathbb{k}$ -algebra and  $A = T_S(V)/(R)$  is quadratic, with  $V$  finite-dimensional and  $R \subset V \otimes_S V$  an  $S$ -sub-bimodule. The quadratic dual records which linear functionals on products of two generators vanish on the relations. Proposition 3.2 makes the exchange explicit: its degree 2 component is the  $S$ -dual of the relation space of  $A$ . Recall that  $V^* = \text{Hom}_S(V, S)$  uses right  $S$ -linear maps, whereas  ${}^*V = \text{Hom}_{S^{\text{op}}}(V, S)$  uses left  $S$ -linear maps. The tensor evaluations in (2.3) are

$$\begin{aligned}\theta_r(g \otimes f)(v \otimes w) &= g(f(v)w), \quad f, g \in V^*, \\ \theta_\ell(g \otimes f)(v \otimes w) &= f(vg(w)), \quad f, g \in {}^*V.\end{aligned}\tag{3.1}$$

Both reverse the order of the factors. The quadratic dual replaces  $V$  by an  $S$ -dual and  $R$  by the maps vanishing on  $R$ .

**Convention.** We identify  $V^* \otimes_S V^*$  with  $(V \otimes_S V)^*$  via  $\theta_r$ , and  ${}^*V \otimes_S {}^*V$  with  ${}^*(V \otimes_S V)$  via  $\theta_\ell$ . Thus  $R^\perp$  and  ${}^\perp R$  in tensor algebras mean the inverse images of the orthogonal complements in Definition 2.28 under these evaluations.

**Definition 3.1** ([BGS96, §2.8]). The *right quadratic dual* and *left quadratic dual* of  $A$  are

$$\begin{aligned}A^! &:= T_S(V^*)/(R^\perp), \quad R^\perp = \{F \in V^* \otimes_S V^* \mid \theta_r(F)(R) = 0\}, \\ {}^!A &:= T_S({}^*V)/({}^\perp R), \quad {}^\perp R = \{F \in {}^*V \otimes_S {}^*V \mid \theta_\ell(F)(R) = 0\}.\end{aligned}\tag{3.2}$$

The generators  $V^*$  and  ${}^*V$  have degree 1.

We use  $A^!$  unless a side is specified. The construction uses the ordinary  $S$ -dual of  $V$ , placed in degree 1, not the graded  $S$ -dual  $V^{(*)}$ , which has degree  $-1$ . It is independent of a chosen presentation:  $V = A_1$  and  $R = \text{Ker}(A_1 \otimes_S A_1 \rightarrow A_2)$  are intrinsic to  $A$ . An isomorphism *over  $S$*  means one restricting to  $\text{id}_S$  on degree 0.

**Proposition 3.2.** The algebra  $A^!$  is quadratic and

$$A_0^! = S, \quad A_1^! = V^*, \quad A_2^! \cong R^*.$$

In particular,  $\dim_{\mathbb{k}} A_2^! = \dim_{\mathbb{k}} R$ .

**Proof** Restriction to  $R$  gives an exact sequence of  $S$ -bimodules

$$0 \longrightarrow R^\perp \longrightarrow V^* \otimes_S V^* \xrightarrow{F \mapsto \theta_r(F)|_R} R^* \longrightarrow 0.\tag{3.3}$$

Surjectivity follows from Theorem 1.9:  $R$  has a complement as a right  $S$ -module, so a right  $S$ -linear map  $R \rightarrow S$  extends by zero on that complement. Now use Proposition 2.14; the dimension statement follows from Proposition 1.43.  $\square$

Adding relations has the opposite effect after dualising. For  $S$ -sub-bimodules  $R \subset R' \subset V \otimes_S V$ ,

$$R \subset R' \implies (R')^\perp \subset R^\perp \implies (T_S(V)/(R'))^! \twoheadrightarrow (T_S(V)/(R))^!.$$

#### 3.2 Recovering the original algebra

**Lemma 3.3.** Let  $U \subset M$  be an  $S$ -sub-bimodule of a finite-dimensional  $S$ -bimodule. Under the evaluation isomorphisms of Lemma 1.49,

$${}^\perp(U^\perp) = U, \quad ({}^\perp U)^\perp = U.$$

**Proof** Every  $u \in U$  is annihilated by every  $f \in U^\perp$ , so  $U \subset {}^\perp(U^\perp)$ . If  $m \notin U$ , then  $m + U \neq 0$  in  $M/U$ . By Corollary 1.10,  $M/U$  embeds as a direct summand of some  $S^{\oplus n}$  as a right module. At least one coordinate of the image of  $m + U$  is non-zero, giving a right  $S$ -linear map  $M/U \rightarrow S$  non-zero on  $m + U$ . Its composite with  $M \rightarrow M/U$  belongs to  $U^\perp$  and does not vanish on  $m$ . Hence  $m \notin {}^\perp(U^\perp)$ . The proof of  $({}^\perp U)^\perp = U$  uses left modules instead.  $\square$

**Theorem 3.4.** *Left and right quadratic duality are canonically inverse:*

$${}^!(A^!) \cong A \cong ({}^!A)^! \quad \text{as graded algebras over } S.$$

**Proof** *Generators.* Write  $\text{ev}_V : V \rightarrow {}^*(V^*)$ ,  $\text{ev}_V(v)(f) = f(v)$ , for the isomorphism in Lemma 1.49. It identifies the generators of  $A$  and  ${}^!(A^!)$ .

*Relations.* We must check that the induced identification on tensors agrees with evaluation on  $V \otimes_S V$ . The square

$$\begin{array}{ccc} V \otimes_S V & \xrightarrow{\text{ev}_V \otimes \text{ev}_V} & {}^*(V^*) \otimes_S {}^*(V^*) \\ \text{ev}_V \otimes_S V \downarrow & & \downarrow \theta_\ell \\ {}^*((V \otimes_S V)^*) & \xrightarrow{F \mapsto F \circ \theta_r} & {}^*(V^* \otimes_S V^*) \end{array} \quad (3.4)$$

commutes, where the right vertical map is (3.1) applied to the bimodule  $V^*$ . Indeed, for  $v, w \in V$  and  $f, g \in V^*$ , the right  $S$ -action on  $V^*$  in (1.8) gives

$$\begin{aligned} \theta_\ell(\text{ev}_V(v) \otimes \text{ev}_V(w))(g \otimes f) &= \text{ev}_V(w)(gf(v)) \\ &= g(f(v)w) = \theta_r(g \otimes f)(v \otimes w). \end{aligned}$$

Thus, using the identifications (3.1), the relations  ${}^\perp(R^\perp)$  of  ${}^!(A^!)$  correspond to the double annihilator of  $R$  in  $V \otimes_S V$ . Lemma 3.3 identifies that annihilator with  $R$ , so (3.2) gives  ${}^!(A^!) \cong A$ . Interchanging left and right gives  $({}^!A)^! \cong A$ .  $\square$

**Proposition 3.5.** *A choice of symmetrising form  $\tau : S \rightarrow \mathbb{k}$  gives isomorphisms  $A^! \cong {}^!A$  and  $(A^!)^! \cong A$  of graded algebras over  $S$ .*

**Proof** The maps  $\alpha_V(f) = \tau \circ f$  and  $\beta_V(g) = \tau \circ g$  in (1.9) give a bimodule isomorphism  $\beta_V^{-1}\alpha_V : V^* \rightarrow {}^*V$ . For  $f, g \in V^*$ , write  $f' = \beta_V^{-1}\alpha_V(f)$  and  $g' = \beta_V^{-1}\alpha_V(g)$ . Then

$$\begin{aligned} \tau(f'(vg'(w))) &= \tau(f(v)g'(w)) = \tau(g'(f(v)w)) \\ &= \tau(g(f(v)w)) \quad (v, w \in V). \end{aligned}$$

By (2.11),  $(\beta_V^{-1}\alpha_V)^{\otimes_S 2}$  therefore identifies  $R^\perp$  with  ${}^\perp R$ . It induces  $A^! \cong {}^!A$  on the presentations (3.2). Apply the same comparison to  $A^!$  and then Theorem 3.4 to obtain  $(A^!)^! \cong {}^!(A^!) \cong A$ .  $\square$

Theorem 3.4 needs no choice of form. Proposition 3.5 permits the usual statement that quadratic duality is an involution, but the identification of left and right duals uses  $\tau$ .

### 3.3 Tensor algebras and monomial relations

**Example 3.6.** *Taking  $R = 0$  or  $R = V \otimes_S V$  in (3.2) gives*

$$\begin{aligned} (T_S(V))^! &= T_S(V^*)/(V^* \otimes_S V^*) \cong S \oplus V^*, \\ (T_S(V)/(V \otimes_S V))^! &= T_S(V^*). \end{aligned}$$

*In  $S \oplus V^*$ ,  $(V^*)^2 = 0$  and  $(s, f)(t, g) = (st, sg + ft)$ . Thus a tensor algebra with no relations is dual to an algebra in which every product of two positive degree elements is zero.*

For the rest of this subsection, take  $S = \bigoplus_{i=1}^r \mathbb{k}e_i$ , with  $e_i e_j = \delta_{ij} e_i$  and  $\sum_i e_i = 1$ . Choose a basis  $\mathcal{B}_{ij}$  of each  $e_i V e_j$  and let  $\mathcal{B} = \bigcup_{i,j} \mathcal{B}_{ij}$ . A non-empty word  $b_1 \cdots b_d$  in  $\mathcal{B}$  is *composable* if its adjacent idempotents match. By Lemma 2.8, these words form a basis of the positive degree part of  $T_S(V)$ .

**Definition 3.7.** *The presentation  $A = T_S(V)/(R)$  is *monomial quadratic* with respect to  $\mathcal{B}$  if  $R$  is spanned by a subset  $\mathcal{R}$  of the composable words of length 2 in  $\mathcal{B}$ .*

For  $b \in \mathcal{B}_{ij}$ , define  $b^* \in V^*$  by

$$b^*(c) = \delta_{bc} e_j \quad (c \in \mathcal{B}), \quad e_j b^* e_i = b^*. \quad (3.5)$$

The elements  $b^*$  form a basis of  $V^*$ ; their left and right idempotents are reversed.

**Proposition 3.8.** *For a monomial quadratic presentation:*

- (1) *A basis of  $A$  consists of the  $e_i$  and the non-empty composable words containing no adjacent pair from  $\mathcal{R}$ .*
- (2) *The quadratic dual is monomial, with*

$$R^\perp = \text{span}_{\mathbb{k}}\{c^* \otimes_S b^* \mid bc \text{ is composable and } bc \notin \mathcal{R}\}. \quad (3.6)$$

**Proof** The ideal generated by  $\mathcal{R}$  is spanned by the composable words containing an element of  $\mathcal{R}$  as an adjacent pair. Removing those basis words proves (1). For  $b \in \mathcal{B}_{ij}$  and  $c \in \mathcal{B}_{j\ell}$ , (3.1) and (3.5) give

$$\theta_r(c^* \otimes b^*)(b \otimes c) = c^*(e_j c) = e_\ell.$$

The functional vanishes on every other basis tensor. By Lemma 2.8, the composable  $c^* \otimes b^*$  form a basis of  $V^* \otimes_S V^*$ . Evaluating a linear combination on each  $b \otimes c \in \mathcal{R}$  forces precisely the coefficient of  $c^* \otimes b^*$  to be zero, since  $e_\ell \neq 0$ . Thus the annihilator is the span of the complementary basis tensors, proving (2).  $\square$

**Example 3.9.** *For  $A = \mathbb{k}A_n$ , take  $S = \bigoplus_i \mathbb{k}e_{ii}$  and  $V = \bigoplus_{i=1}^{n-1} \mathbb{k}e_{i,i+1}$  as in Example 2.15. There are no quadratic relations, so  $A^\dagger$  has the reversed generators  $e_{i,i+1}^* \in e_{i+1,i+1} V^* e_{ii}$  and every composable product of two generators is zero. Consequently  $h_{A^\dagger}(t) = n + (n-1)t$ . For  $n = 3$ ,*

$$(\mathbb{k}A_3)^\dagger \cong (\mathbb{k}A_3/(e_{13}))^{\text{op}}, \quad (\mathbb{k}A_3/(e_{13}))^\dagger \cong (\mathbb{k}A_3)^{\text{op}}.$$

*These are graded isomorphisms fixing the diagonal idempotents.*

**Example 3.10.** *Take  $A = T_S(V)/(ab)$  from Example 2.17, where  $S = \mathbb{k}e_1 \oplus \mathbb{k}e_2$ ,  $e_1 a e_2 = a$ , and  $e_2 b e_1 = b$ . The only composable words of length 2 are  $ab$  and  $ba$ . Thus*

$$A^\dagger = T_S(V^*)/(a^* b^*), \quad A^\dagger = S \oplus \mathbb{k}a^* \oplus \mathbb{k}b^* \oplus \mathbb{k}b^* a^*.$$

*The original relation  $ab = 0$  corresponds to the surviving word  $b^* a^*$ ; the surviving word  $ba$  corresponds to the dual relation  $a^* b^* = 0$ . In both algebras the Hilbert series is  $2 + 2t + t^2$ .*

### 3.4 Symmetric and exterior algebras

**Proposition 3.11.** *For a finite-dimensional  $\mathbb{k}$ -vector space  $V$ ,*

$$\text{Sym}(V)^\dagger \cong \bigwedge D(V), \quad (\bigwedge V)^\dagger \cong \text{Sym}(D(V)).$$

**Proof** Here  $S = \mathbb{k}$  and  $V^* = D(V)$ . Choose a basis  $x_1, \dots, x_n$  of  $V$ , with coordinate functionals  $x_1^*, \dots, x_n^*$ . The evaluation (3.1) reads

$$\theta_r(x_j^* \otimes x_i^*)(x_p \otimes x_q) = \delta_{ip} \delta_{jq}.$$

For  $F = \sum_{i,j} c_{ij} x_i^* \otimes x_j^*$ , vanishing on each  $x_p \otimes x_q - x_q \otimes x_p$  is equivalent to  $c_{pq} = c_{qp}$ . Thus the orthogonal complement of the symmetric relations is spanned by  $(x_i^*)^2$  and  $x_i^* x_j^* + x_j^* x_i^*$  for  $i < j$ , which are the exterior relations. Vanishing on the exterior relations instead gives  $c_{ii} = 0$  and  $c_{pq} + c_{qp} = 0$ . The orthogonal complement is therefore spanned by  $x_i^* x_j^* - x_j^* x_i^*$  for  $i < j$ . Now use the presentations in (2.10).  $\square$

The proof works in characteristic 2: the relations  $(x_i^*)^2 = 0$  remain part of the exterior presentation even when the signs  $+$  and  $-$  coincide. Example 2.29 is the calculation for two generators.

### 3.5 Direct sums

For graded algebras  $A, B$ , the finite direct sum  $A \oplus B$  has componentwise multiplication, unit  $(1_A, 1_B)$ , and  $(A \oplus B)_i = A_i \oplus B_i$ . In particular, its degree zero algebra is  $A_0 \oplus B_0$ .

**Proposition 3.12.** *Let  $A = T_S(V)/(R)$  and  $B = T_T(W)/(Q)$  be quadratic algebras. Then  $A \oplus B$  is quadratic over  $S \oplus T$ , and*

$$(A \oplus B)^! \cong A^! \oplus B^!.$$

**Proof** Regard  $V$  and  $W$  as  $(S \oplus T)$ -bimodules supported on the two respective factors. Mixed tensors vanish, so

$$\begin{aligned} T_{S \oplus T}(V \oplus W) &\cong T_S(V) \oplus T_T(W), \\ (V \oplus W) \otimes_{S \oplus T} (V \oplus W) &\cong (V \otimes_S V) \oplus (W \otimes_T W). \end{aligned}$$

The relations are  $R \oplus Q$ . Right module homomorphisms also split over the two factors, and their annihilator is  $R^\perp \oplus Q^\perp$ . Apply Definition 3.1.  $\square$

For example,  $(\mathbb{k}A_2 \oplus \mathbb{k}A_2)^! \cong (\mathbb{k}A_2)^{\text{op}} \oplus (\mathbb{k}A_2)^{\text{op}}$ . Both degree zero algebras are  $\mathbb{k}^{\oplus 4}$ ; there are no products connecting the two summands.

### 3.6 Tensor products and signs

**Definition 3.13.** *The **ordinary tensor product** of graded  $\mathbb{k}$ -algebras  $A, B$  has underlying vector space  $A \otimes_{\mathbb{k}} B$ , the total grading of Definition 2.24, and multiplication*

$$(a \otimes b)(a' \otimes b') = aa' \otimes bb'.$$

*The **graded tensor product**  $A \otimes_{\mathbb{k}}^{\text{gr}} B$  has the same graded vector space, with multiplication*

$$(a \otimes b)(a' \otimes b') = (-1)^{\deg b \deg a'} aa' \otimes bb' \quad \text{for homogeneous } a, a', b, b'. \quad (3.7)$$

In the ordinary tensor product,  $a \otimes 1$  and  $1 \otimes b$  commute. In the graded tensor product they satisfy

$$(1 \otimes b)(a \otimes 1) = (-1)^{\deg a \deg b} (a \otimes 1)(1 \otimes b).$$

The sign does not change the underlying graded vector space, so both tensor products have Hilbert series  $h_A(t)h_B(t)$  by Lemma 2.25.

**Proposition 3.14.** *Let  $A = T_S(V)/(R)$  and  $B = T_T(W)/(Q)$  be quadratic algebras, and assume that  $S \otimes_{\mathbb{k}} T$  is semisimple. Then both tensor products are quadratic over  $S \otimes_{\mathbb{k}} T$ , and*

$$(A \otimes_{\mathbb{k}} B)^! \cong A^! \otimes_{\mathbb{k}}^{\text{gr}} B^!, \quad (A \otimes_{\mathbb{k}}^{\text{gr}} B)^! \cong A^! \otimes_{\mathbb{k}} B^!. \quad (3.8)$$

The assumption on  $S \otimes_{\mathbb{k}} T$  holds, for example, when both  $S$  and  $T$  are products of matrix algebras over  $\mathbb{k}$ . Over a general field it does not follow from semisimplicity of  $S$  and  $T$  alone.

**Proof** The degree 1 part of  $A \otimes_{\mathbb{k}} B$  is  $(V \otimes_{\mathbb{k}} T) \oplus (S \otimes_{\mathbb{k}} W)$ . By Lemma 2.2, its tensor square over  $S \otimes_{\mathbb{k}} T$  decomposes into the four summands

$$(V \otimes_S V) \otimes_{\mathbb{k}} T \oplus \underbrace{V \otimes_{\mathbb{k}} W}_{\text{order } VW} \oplus \underbrace{V \otimes_{\mathbb{k}} W}_{\text{order } WV} \oplus S \otimes_{\mathbb{k}} (W \otimes_T W).$$

Its quadratic relations consist of  $R \otimes_{\mathbb{k}} T$ ,  $S \otimes_{\mathbb{k}} Q$ , and the mixed relations

$$(v \otimes 1)(1 \otimes w) - (1 \otimes w)(v \otimes 1) \quad (v \in V, w \in W). \quad (3.9)$$

Here the products are taken in the tensor algebra on the degree 1 part, and their span is an  $(S \otimes_{\mathbb{k}} T)$ -sub-bimodule. Let  $C$  be the quotient of that tensor algebra by the three specified relation spaces. The generator maps induce algebra homomorphisms  $A \rightarrow C$ ,  $a \mapsto \bar{a}$ , and  $B \rightarrow C$ ,  $b \mapsto \bar{b}$ . Their images commute: the degree 0 actions already commute across the two factors, and (3.9) makes the degree 1 generators commute. Consequently  $A \otimes_{\mathbb{k}} B \rightarrow C$ ,  $a \otimes b \mapsto \bar{a} \bar{b}$ , is an algebra homomorphism. It is inverse to the canonical homomorphism  $C \rightarrow A \otimes_{\mathbb{k}} B$ , since both composites fix the generating copies of  $S, T, V, W$ . Thus the stated relations give exactly  $A \otimes_{\mathbb{k}} B$ . For the graded tensor product, the same construction uses  $\bar{b} \bar{a} = (-1)^{\deg a \deg b} \bar{a} \bar{b}$  for homogeneous  $a, b$ .

The right  $(S \otimes_{\mathbb{k}} T)$ -duals of the two generator summands are  $V^* \otimes_{\mathbb{k}} T$  and  $S \otimes_{\mathbb{k}} W^*$ , with evaluations

$$(f \otimes t)(v \otimes t') = f(v) \otimes tt', \quad (s \otimes g)(s' \otimes w) = ss' \otimes g(w).$$

These identifications hold for finite free right modules and hence for their direct summands. The reversed tensor evaluation pairs each pure summand with its corresponding dual summand and exchanges the two mixed summands; all other pairings are zero. The pure relation spaces have orthogonal complements  $R^{\perp} \otimes_{\mathbb{k}} T$  and  $S \otimes_{\mathbb{k}} Q^{\perp}$  in the corresponding pure summands. The mixed relation space is the set of pairs  $(z, -z)$  in the two copies of  $V \otimes_{\mathbb{k}} W$ . For mixed tensors, the two reversed evaluations of  $f \in V^*$  and  $g \in W^*$  both give  $f(v) \otimes g(w)$  on the corresponding copy. Thus an annihilating pair of functionals must have equal components. As these simple tensors span the full dual, the mixed annihilator is spanned by

$$(f \otimes 1)(1 \otimes g) + (1 \otimes g)(f \otimes 1) \quad (f \in V^*, g \in W^*).$$

These are the mixed relations of  $A^! \otimes_{\mathbb{k}}^{\text{gr}} B^!$ , proving the first isomorphism in (3.8). For  $A \otimes_{\mathbb{k}}^{\text{gr}} B$ , the minus sign in (3.9) is a plus sign. Its annihilator has a minus sign, giving the ordinary tensor product on the dual side.  $\square$

**Example 3.15.** Take  $A = \mathbb{k}A_2 \otimes_{\mathbb{k}} \mathbb{k}A_2$ . Its degree zero algebra has basis  $e_{ii} \otimes e_{jj}$  for  $i, j \in \{1, 2\}$ , and its degree 1 part has basis

$$a = e_{12} \otimes e_{11}, \quad b = e_{22} \otimes e_{12}, \quad c = e_{11} \otimes e_{12}, \quad d = e_{12} \otimes e_{22}.$$

By Lemma 2.8, the tensor square of the degree 1 part has basis  $ab, cd$ , and both map to  $e_{12} \otimes e_{12}$ . Thus the relation space is  $\mathbb{k}(ab - cd)$ . Using the reversed generators from (3.5), its orthogonal complement is  $\mathbb{k}(b^* a^* + d^* c^*)$ . Hence the dual presentation has an anticommuting relation:

$$ab = cd \text{ in } A, \quad b^* a^* = -d^* c^* \text{ in } A^!, \quad h_A(t) = h_{A^!}(t) = 4 + 4t + t^2.$$

The signs agree with Proposition 3.14; in characteristic 2, the same presentations remain valid with  $-1 = 1$ . Here replacing  $d^*$  by  $-d^*$  removes the minus sign. In this example the sign distinguishes the specified presentations, not the isomorphism classes of algebras.

**Example 3.16.** For  $V = \mathbb{k}x \oplus \mathbb{k}y$ , Proposition 3.14 gives

$$\mathrm{Sym}(V)^! \cong (\mathbb{k}[x] \otimes_{\mathbb{k}} \mathbb{k}[y])^! \cong \frac{\mathbb{k}[x^*]}{(x^*)^2} \otimes_{\mathbb{k}}^{\mathrm{gr}} \frac{\mathbb{k}[y^*]}{(y^*)^2} \cong \bigwedge D(V).$$

If  $\mathrm{char} \mathbb{k} \neq 2$ , the ordinary tensor product of the two square-zero algebras is commutative, whereas the graded tensor product is not:  $x^*y^* = -y^*x^* \neq 0$ . They are therefore not isomorphic. This sign cannot be removed by a change of generators.

### 3.7 Dimensions and the later homological calculation

Equality of Hilbert series in Examples 3.10 and 3.15 is not a general property of quadratic duality; the symmetric and exterior series in Example 2.23 already differ. With several idempotents, it is useful to retain the dimensions of the individual blocks.

**Definition 3.17.** For a locally finite graded algebra with  $A_0 = \bigoplus_{i=1}^r \mathbb{k}e_i$ , its *matrix Hilbert series* is

$$H_A(t)_{ij} := \sum_{d \geq 0} \dim_{\mathbb{k}}(e_i A_d e_j) t^d \quad (1 \leq i, j \leq r).$$

The scalar series  $h_A(t)$  is the sum of all entries of  $H_A(t)$ . The matrix version remembers the left and right actions: (2.6) turns tensor products over  $A_0$  into matrix multiplication, while  $S$ -duality transposes the indices. For  $A = \mathbb{k}A_2$ , for example,

$$H_A(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad H_{A^!}(-t)^{\mathrm{T}} = \begin{pmatrix} 1 & -t \\ 0 & 1 \end{pmatrix}, \quad H_A(t)H_{A^!}(-t)^{\mathrm{T}} = I_2.$$

In contrast,  $h_A(t)h_{A^!}(-t) = 4 - t^2$ , not 1.

The condition called *Koszul* will be defined in Lecture 5 using graded projective resolutions of  $S$ . For such algebras, Lecture 8 will prove the identity

$$H_A(t)H_{A^!}(-t)^{\mathrm{T}} = I_r \quad \text{if } A_0 = \bigoplus_{i=1}^r \mathbb{k}e_i; \quad (3.10)$$

see [BGS96, §2.11]. It is not asserted for every quadratic algebra. For the polynomial and exterior pair, the scalar instance is  $(1-t)^{-n}(1-t)^n = 1$ .

The purpose of constructing  $A^!$  is homological: for a Koszul algebra, its degree  $i$  part computes the space  $\mathrm{Ext}_A^i(S, S)$  of extensions, introduced in Lecture 6, and multiplication corresponds to joining extensions. With right modules and composition taking the right factor first, Lecture 7 will identify the resulting algebra  $\bigoplus_{i \geq 0} \mathrm{Ext}_A^i(S, S)$  with the right quadratic dual  $A^!$  over  $S$ . The choice of right  $S$ -dual is reflected already in degree 1, where  $A_1^* = \mathrm{Hom}_S(A_1, S)$  appears. No homological claim is needed for the calculations in this lecture.

### 3.8 Exercises

**Exercise 3.18.** For  $A = \mathbb{k}A_4/(e_{13})$ , write a presentation of  $A^!$  using reversed generators and find homogeneous bases of  $A$  and  $A^!$ . Check the double dual using Proposition 3.8. Hint. Among  $e_{12}e_{23}$  and  $e_{23}e_{34}$ , only the first is a relation. Complement the forbidden pairs and reverse their order.

**Exercise 3.19.** Let  $S = \mathbb{k}e_1 \oplus \mathbb{k}e_2$  and let  $V$  have basis  $a, b$  with  $e_1 a e_2 = a$  and  $e_2 b e_1 = b$ . Compute the quadratic dual of  $T_S(V)/(ab, ba)$  and the Hilbert series of both algebras. Check. The two series are  $2 + 2t$  and  $2/(1-t)$ .

**Exercise 3.20.** For the four generators in Example 3.15, replace the relation  $ab - cd$  by  $ab - \lambda cd$ , where  $\lambda \in \mathbb{k}$ . Compute the quadratic dual relation for every  $\lambda$ , including  $\lambda = 0$ , and recover the original relation by dualising again. Hint. Evaluate  $\mu b^* a^* + \nu d^* c^*$  on  $ab - \lambda cd$ . For  $\lambda \neq 0$ , rescale  $d$  to compare with Example 3.15.

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